

The Socioeconomic Effects of Forced Displacement: Evidence from the Tennessee Valley Authority

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September 2024

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Abstract

We examine the socioeconomic effects of forced migration by focusing on individuals who were displaced by the Tennessee Valley Authority (TVA) dam projects in the 1930s. We use data from the relocation program associated with the TVA and link it to US Census data. We compare individuals who were impacted by the dam-induced flooding with individuals in the same counties that were not affected by the dam construction due to their proximity to the Tennessee River. We find evidence that individuals who were impacted by the dam projects are more likely to participate in the labor force, with an influx into unskilled occupations. They are also more likely to pay higher rent prices conditional on renting. We examine racial disparities in outcomes and find that, after the relocation, Black men are more likely to be employed in unskilled occupations compared to White families and White men.

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1. Introduction

Forced displacement resulting from wars, natural disasters, and human rights violations has become increasingly common. The United Nations estimates the number of forced displaced individuals around the world to reach 120 million by the end of 2024. Understanding the consequences of forced displacement is therefore becoming increasingly important. Its severe impacts can likely persist long after the incident. Previous studies have shown evidence of negative impacts of relocation driven by natural disasters on labor market outcomes (Mueller and Quisumbing, 2009) and financial resources (Chen, 2020). Other studies have examined the impacts of the Great Migration on African Americans in the US (Tolnay, 2003). There is little evidence on short- and long-term impacts of forced displacement induced by government development projects, although this practice of eminent domain is very common in the United States.

This paper investigates the effects of forced migration by leveraging a natural experiment from the 1930s - the Tennessee Valley Authority (TVA) dam projects. The TVA created dams in seven states in the southeastern United States as part of the New Deal and displaced thousands of people. We use data on the displaced individuals, which we have matched to US Census records using a probabilistic matching algorithm, to examine the effects of forced displacement on a range of medium-run socioeconomic outcomes, including labor market participation and household status. Development of these reservoirs was an effort to combat the lack of economic opportunity in the region, especially after the downturn of the Great Depression. However, the success of this New Deal program on the people it displaced is not widely understood.

The Tennessee Valley Authority Act of 1933 marked the first time a government agency had power to govern the development of an entire region. Not only did the act improve infrastructure and rural life, it drastically improved the region's standard of living within a relatively brief period (Walker 1998). "The T.V.A. has relocated over 12,000 farm families, moved entire villages, helped whole communities plan a better future, and withal it has demonstrated that social planning can win the enthusiastic support of the people at the muzzle end of it. So the T.V.A. has set a pattern for social planning for a generation" (Carr 1948). However, historical research suggest the TVA perpetuated segregation and discrimination against Black individuals (Jones 2016, Alderman and Brown 2010). For instance, using TVA's relocation records and oral histories, Walker (1998) argues the Authority's benefits were not fairly distributed. The agency maintained a policy of institutionalized segregation and discrimination against Black people which also prevented them from deriving direct benefits from TVA's employment and educational programs. Walker (1998) also states that the beneficial programs provided by the TVA favored prosperous landowners over poorer landowners and tenants of both races. Though relocation workers attested to abiding by TVA's removal worker's code of ethics, many assumed African Americans did not have a sense of community, hence only using the word "community" in documentation towards White families, suggesting TVA workers were less sympathetic about removing Black families from their communities than for White families. A chief criticism of the dam construction project is the lack of a systematic follow up of the removed families to their new locations (Brown 1951). Brown states, "If this was done, a better basis would be available for judging the success and failures of family removal and relocation in the Tennessee Valley."

We use a difference-in-differences design comparing individuals who were relocated as part of the TVA dam construction with individuals who were not relocated. We use data from the TVA's case files on relocated individuals, which we combine with US Census records from 1900-1940 to examine effects on socioeconomic well-being. Our comparison group includes individuals living in the same counties as those forced to relocate, but were not in the path of the dam construction. As a robustness check, we consider two more control groups, the first being Georgia residents who would be displaced 5-7 years after those in Alabama due to the rollout timing of the dam construction. This comparison group serves as a have-not-yet-been-treated group. For the second comparison group, we match control to treated counties based on 1930 characteristics. Therefore, the control group consists of people living in similar counties as those who were forced to relocate. We find largely similar results for all three control group specifications

Our findings indicate that displaced families were more likely to pay higher rent prices, and displaced men were more likely to be in the labor force, be employed, and work in unskilled labor. We also find suggestive evidence that displaced men were less likely to work skilled jobs. To examine the robustness of our main findings, we use a control group of counties that are similar in baseline characteristics to the counties where displaced individuals were living, and find largely similar results. These results suggest that while the forced relocation improved labor market outcomes, it came at the cost of living in places with higher rents and working in less skilled jobs.

In addition, we examine whether there are differential impacts of forced relocation by race, to determine whether discrimination played a role in the effects of forced relocation. Our evidence suggests that discrimination against Black men was likely part of the relocation process, as Black families saw larger increases in rent prices than White families. Black men were also significantly more likely to participate in the labor force than White men, but were not more likely than white men to be employed. We also find that Black men were more likely than White men to work in unskilled jobs, but were not less likely than White men to work skilled jobs. These findings suggest that Black individuals faced some discrimination as a result of the TVA's relocation program.

This study contributes to different strands of the literature. First, it adds to the body of work that examines the effects of displacement through the lens of the Great Migration in the United States. Although the identification strategy in these studies is affected by the fact that African Americans made a decision to migrate north for better opportunities, it has been argued that harsh living conditions left many with little choice but to relocate. The existing evidence shows that relocation had positive impacts: it increased earnings of Black migrants (Collins and Wanamaker, 2014) and schooling for migrant children, which decreased the White-Black education gap (Baran et. al, 2024). However, Derenoncourt (2019) finds that destination cities responded to an influx of Black migrants by changing city characteristics - such as increasing policing - thus reducing the benefits of migration. This response to an influx of Black migrants also led to an increase in incarceration rates for Black individuals (Eriksson, 2019). We add to these studies by examining how Black and White people respond similarly and differently to a more short-distanced forced relocation.

We also contribute to the literature on the impact of displacement after natural disasters. For instance, previous studies have shown that Hurricane Katrina had small impacts to victims' wage earnings and income, with labor supply and earnings returning to pre-disaster levels for those who relocated back after

the hurricane (Deryugina et al., 2018, Vigdor, 2007). However, these natural disasters can cause financial hardships, especially for poorer countries and counties with greater income disparity (Johar, 2022; Cavallo, 2022; Lim et al., 2017). The National Research Council (2006) states, “...more recent research suggests that such [positive rebounding from natural disasters] findings paint an overly simplified and perhaps overly optimistic picture of post-disaster recovery.”

Finally, we contribute to the literature on the impact of displacement caused by government development programs. For instance, Chyn (2018) and Chetty et al (2016) examine the impact of the Moving to Opportunity program which led to the demolition of public housing and provided housing vouchers to move people from high-to low-poverty neighborhoods in the 1990s. They show this form of displacement improves economic outcomes for children who moved. These studies raise the question of whether providing an opportunity to move, rather than forcing relocation, is better for citizens in that area. Chyn (2018)’s results suggest that forcing relocation leads to beneficial longer-run outcomes for children, perhaps because short-term constraints prevented such moves when they weren’t forced.

We add to the ‘New Deal’ literature by examining the effects of its establishment. Many New Deal programs were shown to offset the repercussions of the Great Depression (Fishback et al, 2007). Fishback et al saw that an increase in relief spending from these New Deal programs led to a decline in infant mortality, suicide rates, and deaths from infectious diseases. Majority of the grants and loan programs were geared towards the poor and unemployed and raised income, increased property values near new public housing projects, while also reducing property crime rates (Fishback and Kachanovskaya, 2016; Fishback and Kollmann, 2015; Fishback et al, 2010). However, certain programs also led to a reduction in employment and labor force participation, especially for women (Taylor, 2011; Sundstrom, 2001).

The Tennessee Valley Authority has not been examined in great depth within economics. The TVA reservoirs led to a temporary increase in malaria rates, later reduced by TVA’s anti-malaria programs (Kitchens, 2013). Its electrification had no significant effects on the productivity of farming or manufacturing (Kitchens, 2014). However, Kline and Moretti (2014) find that the growth rate of manufacturing employment was significantly larger in TVA counties than non-TVA counties; hence, TVA sped the industrialization of the Tennessee Valley. The authors focus on the economy of TVA counties compared to non-TVA counties, while our paper will focus on the people who were impacted by the TVA’s development.

This project contributes to the literature on forced relocation as well as the New Deal literature while examining outcomes of one of its largest programs closely (Kelly, 2024). We provide the first causal evidence of the effects of the TVA dam project on individual outcomes. Overall, our findings show that forced relocation from Tennessee Valley Authority’s dam construction project improved labor market outcomes, yet increased rent prices as well. The defining aspect of this project is that these events, like extreme weather events, are forced and citizens do not have a choice of whether to stay or leave their homes.

The paper is organized as follows. Section 2 provides background information on the creation of the Tennessee Valley Authority and its relocation program. Section 3 describes the data sources used to identify forced relocation effects. It also contains an explanation of our linking process to historical

census. Our empirical approach is walked through in Section 4, followed by our conceptual framework in Section 5. Results are presented in Section 6 along with a robustness check and mechanisms in Sections 7 and 8. We conclude in Section 9.

2 Background

2.1 Historical Background: Tennessee Valley Authority and the Great Depression

The 1930s Great Depression was one of the worst economic depressions in history. During the time, many Americans placed blame on President Herbert Hoover for steep unemployment and massive bank failures. His successor, President Franklin D. Roosevelt, began the ‘New Deal’ recovery program to stimulate economic recovery. The Tennessee Valley Authority (TVA) was established through the Tennessee Valley Authority Act of 1933 as part of the New Deal. Its duties included improving the navigability of the Tennessee River; providing flood control through reforestation of marginal lands in the Tennessee Valley watershed; developing agriculture, commerce and industry in the valley; and operating the hydroelectric dams³. The Act also gave TVA the power of eminent domain, allowing them to acquire land along the Tennessee River for developmental reasons. These hydroelectric dams, shown in Figure 1, were constructed to expand the Tennessee River, which runs through the seven states TVA oversees: Alabama, Georgia, Kentucky, Mississippi, North Carolina, Tennessee, and Virginia. Construction of these dams caused numerous towns to be submerged underwater, forcing residents to move. The dams, built along the Tennessee River, roughly move east to west as seen in Figure 1. On average, construction of each dam took about 3 years to complete between 1933 and 1953.

2.2 Tennessee Valley Authority’s Relocation Program

The TVA provided a relocation program to assist families displaced by the dams to move to a new location. TVA sent interviewers to each household that would be affected. These interviewers filled out a relocation application with the family that contained detailed information about their house, land, and demographics for all individuals living there. This information was then used by the TVA to determine suitable dwellings for families to relocate to. Though the TVA did not provide compensation for relocation, it provided its resources to find new homes for sale, appraising them, and relocating families to farms that best suited their needs (Satterfield, 1937, p 258). To our knowledge, a rubric that details how and in which order new dwellings were chosen does not exist. However, it has been documented that families were relocated to a new home and land that was similar to their previous home and land (McCarthy, 1949, p 49; Satterfield, 1937, p 258). Though most families were tenant farmers, families and landlords who owned their homes and land were offered fair market prices (McCarthy, 1949, p 55-56). Relocation was provided for both the living and deceased,⁴ and an estimated 15,000 families and 30,000 graves were relocated across all seven states.

To our knowledge, there were not clear documented rules determining the TVA’s decision to flood particular areas. However, Howard (1936) explains the steps towards dam construction the TVA took in order to implement the program as efficiently as possible. In the twelve step plan towards the project completion, fully interpreted into a list in the Appendix Figure A1, the majority of the steps were focused on the employees hired to construct the dams. These included understanding the labor supply of the

³ Transcript of the Tennessee Valley Authority Act (1933)

⁴ The Tennessee Valley Authority would oblige by the family’s wishes to relocate their loved one’s grave.

region as well as providing resources for those employees such as a camp to house them and resources to educate them on-and off-duty. Other steps focused on cooperation between local governments and the TVA as the region developed and the project progressed. One of the last steps pertained to the relocation of families. As mentioned, detailed facts about families who were in the path of the dam construction were obtained; these details were used by social scientists to analyze those who were being displaced and to determine relocation strategy. As mentioned by Howard, an analysis determining the extent to which [relocated] families have successfully integrated themselves into a new life, without either cultural or economic loss, will be important to document the success of the program. To our knowledge, a full analysis as such has yet to happen.

3. Data

3.1 Information on Relocated Families

We obtained information on relocated families in Alabama from the “U.S., Tennessee Valley, Family Removal and Population Readjustment Case Files, 1934-1953” dataset via ancestry.com. This dataset contains pdf files of original relocation applications which provides rich information about each relocated family in order for the TVA to decide where to move them (Figures 2.1 and 2.2). Housing details such as ownership of property, dwelling characteristics, and farming style of crops were described at the household level. Employment information such as occupation, income, and highest level of education were provided for all members of the family living in the house at the time. For each family member, demographics such as gender, race, birth year, relation to the head of the household, along with their full name⁵ were provided as well. To our knowledge, we are the first to transcribe and digitize pdf files of TVA’s relocation applications for the state of Alabama, which constitutes over 6000 relocated families.

3.2 Census Data

After obtaining information on which families were relocated, we linked those families to the 1930s census by matching on name and other defining characteristics such as birth year, gender, and residing county, to gather our outcome variables. The census also allowed us to gather outcome variables for non-relocated families who were not in the path of the dam construction which constitutes our control group(s). These include a farming indicator, an ownership indicator, monthly rent payments, labor force participation, employment, and occupation score.⁶ We obtain data prior to dam construction from the 1900, 1910, 1920, and 1930 censuses, with post-construction data coming from the 1940 census.

3.3 Linking Relocated Families Data to Census Data

⁵ Full names are encrypted as histids for analysis. We only use full names in the IPUMS census lab to link relocated families to the census to obtain outcomes and to identify them as part of the treated group.

⁶ The census provides an Occupation Score which ranges from 1 to 80 and represents the median total income for each occupation. Higher values align with high end jobs compared to the lower values which align with lower end jobs. Appendix Table A5, shows our results for occupation score outcomes, however we exclude them from our analysis due to incorrect measure of prestige.

We use record linking methods to link the data on relocated families to the 1930 census data using a probabilistic matching algorithm that matches individuals on name, birth year, location, and gender. We require an exact match on county and gender, and allow for flexible matching on both name and age. However, we require that individuals be born in the same decade and they have the same first and last initial. We are able to link about 62% of the relocated families data to the 1930 census. Of the 62% linked relocatees, about 40% were linked to the 1940 census. Lastly, about 15% of all relocatees were linked across previous censuses. The population of linked relocated families contains more males, White individuals, farmers, have higher income and literacy rate on average, and less likely to rent their property compared to those who did not link across earlier censuses, as shown in Table A7. Our final sample consists of families and of men in the state of Alabama who have both 1930s and 1940s census information and linked information for at least one of the 1900s, 1910s, or 1920s census⁷.

4. Empirical Approach

To understand impacts of a forced relocation, we compute simple difference-in-difference models that observe outcomes of those who were forced to relocate, the treated group, compared to those who were similar but not forced to relocate, the control group. We look at the household level for family outcomes such as farming status, renting status, and rent price⁸. For individual labor outcomes such as employment status, labor force status, and switching into unskilled or skilled occupations (Ferrie, 1997), we look solely at men because married women were not consistently asked about employment when censuses were conducted. Therefore women and children are excluded from individual employment outcomes.

In order to measure effects on a forced relocation, we structured our comparison group to consist of people living within the same counties as those treated, but were not forced to relocate due to their proximity to the Tennessee River. Figures 3.1 and 3.2 highlight Alabama areas near Wheeler and Guntersville dams that displaced families and men. Those who were living in the same counties as the displaced people, but not forced to relocate due to proximity to the Tennessee River, comprise our main comparison group. The figures also show our robustness check comparison groups: (i) soon-to-be relocated residents in the state of Georgia (have-not-yet been treated), and (iv) people living in similar counties as those displaced based on matching county characteristics. Our identification strategy suggests that in the absence of the TVA's dam construction project, those forced to relocate would have progressed in property and labor market outcomes similar to those who comprise our control groups and were not forced to relocate. We test this "parallel trends" assumption using Figures 4, 5.1, and 5.2, which shows support for the assumption for all variables in our control groups except Farm and Skilled Occupation. In this setting, we refer to our likelihood of farming and switching into skilled occupation results as a suggestive effect, but not casual.

We estimate two difference-in-difference models to determine the effects of a forced relocation:

⁷ We exclude women for individual outcomes due to (i) most women not working nor (ii) asked about their employment status in census interviews during this time period.

⁸ Rent prices are only reported for non-farm, renter-occupied units as described in the IPUMS database.

$$Y_{it} = \beta_0 + \beta_1 Black_i + \beta_2 Treated_i + \beta_3 Treated_i * Post_t + \tau_t + \varepsilon_{it} \quad (1)$$

Where Y_{it} in Equation 1 represents socioeconomic outcomes for household or individual male i in census year $t \in \{1900, 1910, 1920, 1930, 1940\}$. $Black_i$ is a binary variable⁹ that equals 1 if the race of the family or male is Black and 0 if race is White. $Treated_i$ indicates whether the families or men, i , were displaced by the dam construction. The interaction between $Treated$ and $Post$ indicates a displaced family or man in the post period 1940. α_3 is our coefficient of interests, representing the socioeconomic effects of being displaced by TVA's dam construction projects. Our error term is represented by ε . All regressions use robust standard errors and are run separately for each control group.

Table 1 shows summary statistics for our variables for both individual and household outcomes in 1930, prior to the dam project. Comparing relocated versus non-relocated individuals and families, we show that on average, the treated group are less likely to participate in the labor force and be in skilled jobs, more likely to be in unskilled occupations, are more white, and pay less in rent compared to either control group. Relocated families are, however, more likely to farm, own their property, and have lower housing values and rent prices than non-relocated families.

5. Conceptual Framework

To inform our empirical work, we discuss the possible channels in which our outcomes may respond to a forced relocation. First consider the setting of our study; Alabama is primarily a rural state, hosting many farmers on its land. With the creation of the dam project, the TVA sought workers from the local region that would help with the construction. As shown in the Appendix, Table A1, the steps towards implementation of the dam construction primarily focused on the employees. With an emphasis on the employees and the job availability the dam project would bring, we expect to see an increase in employment in these construction jobs, hence a decrease in farming. Due to Alabama hosting primarily poor tenants in its rural areas, we expect to see differences in outcomes between urban and rural areas. It may be the case that a forced relocation improved outcomes for poor rural tenants, especially by creating more job opportunities. However, if TVA favored prosperous landowners over poorer ones (Walker 1998), then we may expect to see such outcomes improve more so for urban relocatees.¹⁰

As the availability for employment increases, a forced relocation might also provide the opportunity of occupation switching. For example, construction jobs for the dam project is seen as an unskilled occupation, hence we foresee an influx into unskilled occupations whereas we might see a decrease into skilled occupations. This assumption is also strengthened by the low share of Alabama residents having a high school diploma or higher as seen in the Appendix, Table A7.

⁹ We drop nonblack and nonwhite races due to a small amount of observations. There is also no evidence of multiple races per household in our sample during this time period.

¹⁰ We test this assumption in our analysis and find while there is no evidence of selection into treatment by the TVA, there are differences in outcomes between renters and owners, Appendix Tables A9-A11. This could indicate differential effects by socioeconomic status.

Finally, we consider relocation effects on rent prices. From historical literature on the TVA, Satterfield (1937) states landlords either did not repurchase farms or purchased smaller farms which reduced their need for tenants. Scarcity in tenant vacancies could induce competition which can result in increases in rent prices from landlords. We also suspect an increase in rent prices due to an influx of people moving into specified areas, hence shifting the demand curve for housing.

With the TVA Act of 1933 occurring in the midst of the Segregation Era in the South, we expect to see differences in effects between Black and White relocatees. As stated above, we anticipate the dam construction project causing an increase in employment. However, this increase may have favored White men. As rent prices potentially increase due to scarcity in tenant vacancies, White families may experience a “first come, first served” approach in housing availability, leaving Black families last in consideration and hence, left with higher rent prices compared to White families.

6. Results

We first report our findings of the effect of TVA relocation on housing and labor market outcomes in Panel A of Table 2, graphically represented in Figure 6. First, our household outcomes indicate that after a forced relocation, people were less likely to farm. While there was almost no change in renting status overall, rent prices increased for non-farm relocatees by 25.3%. Next, we look at our individual outcomes. There was about a 11% increase in labor force participation as well as a 7% increase in the likelihood of being employed. However, displaced men were 13.6% more likely to work in unskilled occupations and less likely to work in skilled occupations, although small and insignificant, compared to those who were not forced to relocate. Our main results point to the potential of a forced relocation improving labor market outcomes, although these improvements may come at a cost i.e. living in more expensive places and working less skilled jobs.

We also examine whether there is a differential impact of forced relocation by race in Table 3, graphically shown in Figure 7. Overall, both Black and White families and men see outcomes in the same direction as our main results, which indicates an improvement in labor market outcomes and property status, along with an increase in rent prices. We run a triple interaction analysis using race, which helps us identify any difference in outcomes by race. Relative to our within-counties comparison group, there appears to be no significant difference in the decline of the likelihood to farm, skilled occupations, or increase in rent prices for both Black and White families and men. This means both Black and White families and men experienced similar treatment in these areas after relocating. However, we do show a significant difference in renting status and labor force participation. Specifically, Black families were 7% less likely to rent their property and Black men were 17.7% more likely to be in the labor force, compared to a 0.3% decline and a 10% increase for White families and White men respectively. Though there is not a significant difference, we can imply that Black men experience a higher increase in employment status, but also a steeper incline into unskilled occupations. These results suggest differences in the impact of a forced relocation by the TVA between Black and White families/men.

We further understand the effects on those forced to relocate, we examine their occupation. Specifically, we split our unskilled occupation outcome into three categories: unskilled occupations as tested in our

main analysis (Ferrie, 1997), unskilled occupations that do not consist of any jobs relating to farming, and unskilled farming jobs such as farm laborers. This helps us understand if an influx into unskilled occupations is driven by people staying or leaving farming jobs. In the Appendix Table A12, we test all three categories for our main comparison group along with our robustness comparison groups. Column 2 shows an increase into unskilled occupations that does not consist of farming jobs, whereas Column 3 shows a decline in unskilled farm jobs. This is also consistent across our other comparison groups which are described further in the next section. Overall, results suggest that relocatees are leaving farming jobs and moving into other unskilled labor occupations. When looking at these results by race, we find a significant difference between Black and White men into unskilled non-farming jobs, 12 and 5.3 p.p. respectively (Appendix Table A13). Results support our main findings, indicating Black men see a higher increase into unskilled occupations compared to White men, specifically in unskilled non-farming occupations.

A useful feature of the data for the TVA relocation program is that it includes a survey describing the attitude of each household towards the TVA. These responses can help further determine if socioeconomic outcomes can be impacted by a household's viewpoint on the TVA.¹¹ There are five possible responses, ranging from antagonistic to active booster (strongly supportive). We categorized these responses into three categories: Negative, Neutral, and Positive. Our Negative category contains both 'antagonistic' and 'critical' responses, our Neutral category contains the 'neutral' response, and our Positive category contains both 'interested' and 'active booster' responses. Figure 10 shows a graph of families' responses by race. The responses across race are similar: over 61% families were neutral or in favor of the TVA. We then report our estimates for labor market outcomes by each of these categories for both control groups in Table 5. We find that our positive effects on labor force participation and employment are strongest among those who were in favor of or indifferent to the TVA, while those who were against the TVA saw no effect. We see an increase in unskilled occupations for those who lean more in favor towards the TVA, and a decrease in skilled occupations across all attitudes with a larger effect for those strongly against the TVA.

7. Robustness Checks

We formulate a balance test of our sample by testing for selection into treatment by the TVA. In the Appendix, Table A8, we regress treatment status on baseline 1930 characteristics. These characteristics consist of both household level outcomes, such as housing value, renter status, rent price, and farming status, and individual level outcomes such as labor force participation, employment status, and unskilled or skilled occupation status. For our main control group, the likelihood of farming appears to increase one's chance of being in the treated group, along with not being in a skilled occupation. However, most of our sample contains poor farmers so we are not concerned with selection here. Therefore, since most of our results indicate that allocation into treatment is not being driven by these baseline characteristics, we conclude that our sample is balanced.

¹¹ There are reasons to believe that attitude responses towards the TVA may not be truthful as there is not a way to determine the intention behind a household's response (i.e. a family could have selected a response in favor of the TVA in hopes to relocate to a 'nicer' dwelling). Therefore, we proceed with caution in our interpretation of results.

To further test the robustness of our results, we employ an alternate strategy of determining our control groups. Our second control group consists of people living along the Tennessee River in Georgia that will later be displaced by dam construction projects beginning in 1942. We deem this as our have-not-yet been treated, control group. The location of this control group can be seen in Figure 3.1. Using equation (1), we find similar results to our main analysis. In Panel B of Table 2, we find labor market outcomes improved such as an increase in labor force participation and employment, which also led to an increase in unskilled occupations, suggesting a decline in skilled occupations. The likelihood of farming and renting property decreased, however there was a surge in rent prices after relocating. We also created a third control group where we calculate baseline county-level demographics in 1930 and use a nearest neighbor matching algorithm to select the most similar county in Alabama for each county in our treatment group. The counties that were selected in this algorithm can be seen in Figure 3.2. We then estimate the same empirical strategy as before using our matched control group, as seen in Panel C of Table 2 and graphically in Figure 9.2. We find the forced relocation led to an increase in labor market outcomes. The increase in employment led to an increase in unskilled occupations and possibly a decline in farming status; rent prices also increased by 48.9%. Results are similar to our main findings.

When looking at results by race, we see a significant difference between Black and White men's labor outcomes. Black men, while seeing a higher labor force participation, were employed into unskilled occupations more so than White men, Table 4. Suggestively, Black men were also less likely to be in skilled occupations. Table 5 shows that Black families were less likely to rent their property compared to White families in our matched county comparison. These results indicate that our findings are robust and provide further evidence that the TVA relocation program may have facilitated discrimination against Black individuals.

8. Mechanisms

One potential mechanism driving the results is that individuals were relocated to places that were better at baseline before the TVA projects began. To examine whether this is the case, we calculate the average county level characteristics for each county in Alabama in the 1930 Census. We then assign the average county characteristics to each individual in our data based on their county of residence in 1930 and 1940, and estimate the difference-in-differences in Equation 1 for each of the control groups. The results of this analysis are in Tables A1 and A2. From these analyses, we find that individuals displaced by the TVA projects are moved to places that were largely similar to where they were living prior to displacement. However, we do examine relocatees moved to places that were less rural and contained less farmers. This could explain our decline in the farming status findings. We also examine if there is heterogeneity by race in Tables A3 and A4, finding that both Black and White individuals, Black families more so than White families, relocate to places that were less rural and contained less farmers.

Another potential mechanism driving the results is discrimination against Black individuals displaced by the TVA dam projects. Specifically, there is potential evidence of discrimination in rent prices towards Black families and a continuation in systematic segregation with a higher occupancy of unskilled jobs for Black men considering an influx in labor force participation. This evidence can be seen in Tables 3, 4, and 5, where Black individuals see higher rents, are more likely to be in the labor force, yet are more likely to have unskilled jobs. To test whether this is the case, we use nearest neighbor matching to determine the

most similar White individual for each Black individual in both our treatment and control groups to account for baseline differences between Black and White individuals. We match on baseline socioeconomic and demographic characteristics from the 1930 Census, and estimate our differences-in-differences specification for both black and white individuals in Table A6. From this analysis, we see suggestive evidence supporting the existence of discrimination, as Black families experienced larger increases in rent prices compared to White families.. We formally test whether the differences between Black and White individuals in Table A6 are statistically significant, and find a significant difference in farming, suggesting Black families were less likely to move out of farming and into other jobs opportunities that the relocation created. These results provide suggestive evidence that Black individuals displaced by the TVA dam projects faced discrimination.

A third potential mechanism driving the results is that there is differential treatment between renters and homeowners. One potential concern related to this is that individuals were selected into treatment based on their household wealth or other baseline characteristics. We test this in Table A8 by regressing treatment status on baseline characteristics in the 1930 Census, including housing value. We find no evidence that individuals were selected into treatment based on observable characteristics. In Tables A9, A10, and A11, we examine whether there are differential outcomes between renters and owners. We find that owners are less likely to be farmers than renters. Owners are also much less likely to be employed and in the labor force, which leads them to being less likely to be in unskilled or skilled occupations compared to renters. Results indicate that the TVA did not favor homeowners over renters, and -in fact- renters are driving the increase in labor market outcomes.

9. Conclusion

Understanding how forced displacement can affect socioeconomic outcomes is an area of concern for researchers and policymakers. We utilize the natural experiment provided by the forced displacement induced by the TVA dam projects to estimate the causal effect of forced displacement on socioeconomic outcomes. We find evidence that forced displacement improved individuals labor market outcomes, increasing labor force participation and employment. However, we also find evidence that housing costs increased as well, indicating a trade-off between labor market prospects and cost of living.

Another issue of rising importance is racial disparities in socioeconomic outcomes, and whether government programs contribute to these disparities or ameliorate them. Building on a large body of research showing evidence that government programs can exacerbate racial inequality, we find evidence that the TVA had disparate outcomes for Black and White individuals. Specifically, we find that while Black men had larger increases in labor force participation than White men, they also faced a large increase in unskilled occupations and rent prices as well. This indicates that the TVA may have not mitigated racial disparities between Black and White individuals.

Our research points to the vital importance of understanding the impacts that relocation programs have on individuals and how those impacts vary by race. Further research is needed to understand how forced migration can affect where individuals move and whether individuals location drives the disparities we find in our research.

Works Cited

- Alderman, D. H., & Brown, R. N. (2010). When a new deal is actually an old deal: The role of TVA in engineering a Jim Crow racialized landscape. *Engineering Earth: The Impacts of Megaengineering Projects, 1901-1916*.
- Baran, C., Chyn, E., & Stuart, B. A. (2024). The Great Migration and Educational Opportunity. *American Economic Journal: Applied Economics*, 16(3), 354-398.
- Brown, R. G. (1951). Family removal in the Tennessee Valley.
- Carr, Lowell Juilliard. (1948). "Situational Analysis, An Observational Approach to Introductory Sociology" *New York: Harper and Brothers Publishers*, pp. 166-167.
- Cavallo, E. A., Becerra, O., & Acevedo, L. (2022). The impact of natural disasters on economic growth. In *Handbook on the economics of disasters* (pp. 150-192). Edward Elgar Publishing.
- Chen, G. (2020). Assessing the financial impact of natural disasters on local governments. *Public Budgeting & Finance*, 40(1), 22-44.
- Chetty, Raj, Nathaniel Hendren, and Lawrence F. Katz. "The effects of exposure to better neighborhoods on children: New evidence from the moving to opportunity experiment." *American Economic Review* 106.4 (2016): 855-902.
- Chyn, Eric. "Moved to opportunity: The long-run effects of public housing demolition on children." *American Economic Review* 108.10 (2018): 3028-3056.
- Clement, Viviane; Rigaud, Kanta Kumari; de Sherbinin, Alex; Jones, Bryan; Adamo, Susana; Schewe, Jacob; Sadiq, Nian; Shabahat, Elham. (2021). Groundswell Part 2 : Acting on Internal Climate Migration. World Bank, Washington, DC. © World Bank.
<https://openknowledge.worldbank.org/handle/10986/36248> License: CC BY 3.0 IGO.
- Collins, W. J., & Wanamaker, M. H. (2014). Selection and economic gains in the great migration of African Americans: new evidence from linked census data. *American Economic Journal: Applied Economics*, 6(1), 220-252.
- Coussens, C., & Goldman, L. (Eds.). (2007). *Environmental Public Health Impacts of Disasters: Hurricane Katrina: Workshop Summary*. National Academies Press.
- Derenoncourt, E. (2019). *Long-run determinants of US racial inequality: Evidence from the great migration and the FLSA* (Order No. 28235645). Available from ABI/INFORM Collection; ProQuest Dissertations & Theses Global. (2459435355). Retrieved from <http://proxy.library.tamu.edu/login?url=https://www.proquest.com/dissertations-theses/long-run-determinants-us-racial-inequality/docview/2459435355/se-2>

- Deryugina, T., Kawano, L., & Levitt, S. (2018). The economic impact of Hurricane Katrina on its victims: Evidence from individual tax returns. *American Economic Journal: Applied Economics*, 10(2), 202-233.
- Eriksson, K. (2019). Moving North and into jail? The great migration and black incarceration. *Journal of Economic Behavior & Organization*, 159, 526-538.
- Ferrie, J. P. (1997). The entry into the US labor market of antebellum European immigrants, 1840–1860. *Explorations in Economic History*, 34(3), 295-330.
- Fishback, P. V., Haines, M. R., & Kantor, S. (2007). Births, deaths, and New Deal relief during the Great Depression. *The review of economics and statistics*, 89(1), 1-14.
- Fishback, P. V., Johnson, R. S., & Kantor, S. (2010). Striking at the roots of crime: The impact of welfare spending on crime during the great depression. *The Journal of Law and Economics*, 53(4), 715-740.
- Fishback, P. V., & Kollmann, T. (2015). Hedonic Housing Indexes during the Great Depression. *Unpublished manuscript*.
- Fishback, P., & Kachanovskaya, V. (2016). The Politics of the New Deal. *Hoover Institution*, July, 19.
- Fishback, P. (2017). How successful was the New Deal? The microeconomic impact of New Deal spending and lending policies in the 1930s. *Journal of Economic Literature*, 55(4), 1435-1485.
- “Global Trends.” *UNHCR*, www.unhcr.org/global-trends. Accessed 23 Oct. 2024.
- Harrison, A. (Ed.). (1991). *Black exodus: The great migration from the American south*. Univ. Press of Mississippi.
- Howard, T. L. (1936). The Social Scientist in the Tennessee Valley Authority Program. *Social Forces*, 15(1), 29-34.
- Hydroelectric*, www.tva.com/energy/our-power-system/hydroelectric. Accessed 12 Oct. 2024.
- Johar, M., Johnston, D. W., Shields, M. A., Siminski, P., & Stavrunova, O. (2022). The economic impacts of direct natural disaster exposure. *Journal of Economic Behavior & Organization*, 196, 26-39.
- Jones, C. M. (2016). Power for the public good: energy, race and class in the United States.
- Kitchens, C. (2013). A dam problem: TVA's fight against malaria, 1926–1951. *The Journal of Economic History*, 73(3), 694-724.
- Kitchens, C. (2014). The role of publicly provided electricity in economic development: the experience of the Tennessee Valley Authority, 1929–1955. *The Journal of Economic History*, 74(2), 389-419.

- Kelly, Martin. "Top 10 New Deal Programs of the 1930s." ThoughtCo, Sep. 2, 2024, thoughtco.com/top-new-deal-programs-104687.
- Kline, Patrick, & Moretti, Enrico, Local Economic Development, Agglomeration Economies, and the Big Push: 100 Years of Evidence from the Tennessee Valley Authority , *The Quarterly Journal of Economics*, Volume 129, Issue 1, February 2014, Pages 275–331, <https://doi.org/10.1093/qje/qjt034>
- Lim, J., Loveridge, S., Shupp, R., & Skidmore, M. (2017). Double danger in the double wide: Dimensions of poverty, housing quality and tornado impacts. *Regional Science and Urban Economics*, 65, 1-15.
- McCarthy, C. J. (1949). Land acquisition policies and proceedings in tva-a study of the role of land acquisition in regional agency. *Ohio State Law Journal*, 10(1), 46-64.
- Mueller, V., & Quisumbing, A. (2009). Natural disasters and their labor market consequences: evidence from the 1998 flood in Bangladesh. *Working Paper*.
- National Research Council, Division on Earth, Life Studies, Committee on Disaster Research in the Social Sciences, Future Challenges, & Opportunities. (2006). *Facing hazards and disasters: Understanding human dimensions*. National Academies Press.
- Satterfield, M. Harry. "The removal of families from Tennessee Valley Authority reservoir areas." *Soc. F.* 16 (1937): 258.
- Stuart, Bryan A., Taylor, Evan J.; The Effect of Social Connectedness on Crime: Evidence from the Great Migration. *The Review of Economics and Statistics* 2021; 103 (1): 18–33. doi: https://doi.org/10.1162/rest_a_00860
- Steven Ruggles, Sarah Flood, Matthew Sobek, Daniel Backman, Annie Chen, Grace Cooper, Stephanie Richards, Renae Rodgers, and Megan Schouweiler. IPUMS USA: Version 15.0 [dataset]. Minneapolis, MN: IPUMS, 2024. <https://doi.org/10.18128/D010.V15.0>
- Steven Ruggles, Matt A. Nelson, Matthew Sobek, Catherine A. Fitch, Ronald Goeken, J. David Hacker, Evan Roberts, and J. Robert Warren. IPUMS Ancestry Full Count Data: Version 4.0 [dataset]. Minneapolis, MN: IPUMS, 2024. <https://doi.org/10.18128/D014.V4.0>
- “Tennessee Valley Authority Act (1933).” *National Archives and Records Administration*, National Archives and Records Administration, www.archives.gov/milestone-documents/tennessee-valley-authority-act.
- Tolnay, S. E. (2003). The African American “great migration” and beyond. *Annual Review of Sociology*, 29(1), 209-232.

“TVA History - Facts, Timeline & Meaning.” *History.Com*, A&E Television Networks,
www.history.com/topics/great-depression/history-of-the-tva.

Vigdor, J. (2007). The Katrina Effect: Was There a Bright Side to the Evacuation of Greater New Orleans?. *The B.E. Journal of Economic Analysis & Policy*, 7(1).
<https://doi.org/10.2202/1935-1682.1773>

Walker, Melissa. "African Americans and TVA reservoir property removal: Race in a New Deal program." *Agricultural History* 72.2 (1998): 417-428.

Tables and Figures

Figure 1. Map of Tennessee Valley Authority’s dam construction projects

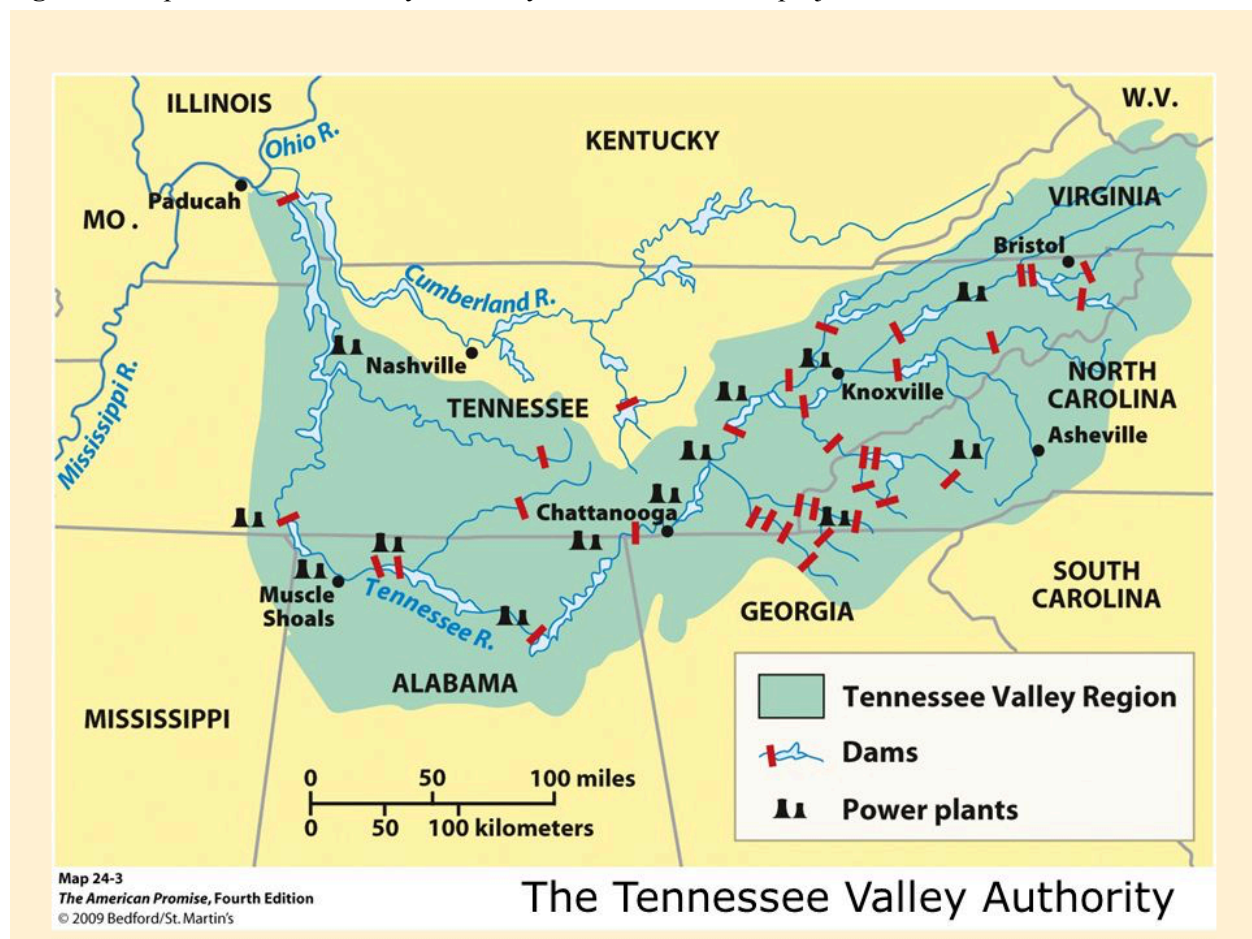


Figure 2.1. Sample Relocation Application

ETA 970
Tennessee Valley Authority 497 Social and Economic Division

Schedule No. 923 INFORMATION Fieldman Stewart
Map _____ County Madison
Date 5/8/35 CONCERNING FAMILIES Civil District _____

1. Name J.C. Malone
2. P. O. address Athens, R#3
3. Highway or road on which home is located _____
4. Miles from home to nearest elementary school _____ Its name _____
5. Miles from home to nearest high school _____ Its name _____
6. Miles from home to nearest church _____ Its name _____
7. Chief trade center for family _____
8. Average number of trips there per month _____ Miles from home _____
9. Years lived at present residence 24 In present community 24 years
10. Information concerning houses lived in during 1934 and 1935:

Year	Owned	No. of Rooms	No. of Sleeping Rooms	Age of House	Electricity	Telephone	Fuel	Plumbing	Condition of House	Condition of Repair
1934										
1935	<u>Own</u>	<u>5</u>	<u>3</u>	<u>35</u>	<u>no</u>	<u>yes</u>	<u>wood</u>	<u>tile</u>	<u>fine</u>	<u>good</u>

11. Source of water supply: Well ✓ Cistern _____ Spring _____
Other _____ Distance from house in back yard.
12. Toilet facilities: Inside _____ Outside ✓ None _____
13. Marital status: M. ✓ S. _____ F. _____ Sep. _____ W. _____ Race White
14. Information concerning husband and wife:

	Age Now	Age When First Married	Age of Wife	Birthplace	Highest School Grade Reached	Condition of Health or Physical Defects
Husband	<u>59</u>	<u>21</u>	<u>1</u>	<u>Kyles Tenn.</u>	<u>8th</u>	<u>Good</u>
Wife	<u>57</u>	<u>19</u>	<u>1</u>	<u>Kyles "</u>	<u>8th</u>	<u>Good</u>

15. No. of children born: Boys 6 Girls 5 No. who have died: Boys 2 Girls 2
16. The newspapers and magazines that the household receives regularly:
Newspapers: Limestone Courier Magazines None
Nashville Tennessean
17. The approximate number of all books in the home 50
18. Denominational preference Baptist Membership in other organizations None
19. Amount of insurance: Life 0 Accident and Health 0
Burial expenses Other 0 Total about \$200
20. Liabilities: Mortgages 0 Other debts 0 Total 0

21. Information concerning members of the family who left home to work after 1920 but later returned and now live at home:

Name	Relation	Type of Work Done	Location of Work	Date Left Home	Date Returned Home	Married While Away
<u>None</u>						

22. Information concerning deaths in the family during past ten years:

Name of Person Who Died	Date of Death	Relation	Residence at Time of Death	Cause of Death	Where Buried
<u>all 10 years or more</u>					

23. Does the family own a car? Yes If so, state the make Ford Model A
Age 5 How many years have they owned it? 6 months
24. Check ownership of the following personal possessions:
Radio _____ Organ _____ Sewing machine ✓ Piano _____
Photograph _____ Other Articles _____
25. The approximate value of furniture:

Room	Value	Room	Value	Room	Value
Kitchen		Living Room		Other Rooms	
Dining room		Bedrooms		Total Furniture	<u>250</u>

26. What is husband's calling or trade? Farming
27. Information on husband's employment during past 10 years (exclude farming):

Dates	Kind of Work Done	Earnings per Month	By Whom and Where Employed	Cause of Change in Employment
<u>None</u>				

28. Kind of work preferred _____
29. Information concerning husband's land (1935):

Location of Land	Tenure	Total Acres	Acres in:				Lay of Soil: Acres		
			Woodland	Pasture	Crope	Other	Level	Rolling	Steep
In	Rents								
Flomago	Rents	<u>120</u>	<u>12</u>	<u>27</u>	<u>71</u>	<u>10</u>		<u>120</u>	
Area	Rents out								
Outside	Rents								
Flomago	Rents	<u>120</u>	<u>12</u>	<u>20</u>	<u>78</u>	<u>10</u>		<u>120</u>	
Area	Rents out								

30. Give location of all land other than home farm, stating whether owned or rented _____
31. Information concerning acres cultivated (1935): Corn 30 Cotton 27
Hay 10 Small grains 4 Sorghum 0 Other crops 0 Total 71
Don share-crops with him

Figure 2.2. Sample Relocation Application Continued.

32. Information concerning type of farming and livestock (1935):

Type of Farming	No. Hogs	No. Pigs	No. Cattle	No. Hogs	No. Sheep	No. Poultry
Cotton	34	4	608	175	5125	345
						6550

33. Type or types of farming preferred Cotton #865

34. If any land is rented, give name of the owner None
Is he related to this family? No If so, give relationship None

35. Is agreement or lease with the owner written or oral? None
When was agreement begun? None When does it end? None

Give information concerning the terms of this agreement for year 1936:

Share of Crops Given to Owner	Cash Payment to Owner	Furnished by Owner
		Livestock Machinery Fertilizer Seed

36. If any land is rented out, give name of the tenant None
Is he related to this family? No If so, give relationship None

37. Is farm covered by a 1934-35 cotton contract? No If so, in whose name? None
Contract No. None No. of acres of cotton allowed in 1934 None No. of acres of cotton planted to government in 1934 None
No. of acres of cultivated land under contract None No. of acres of this land to be flooded None If tenant, no. of acres in cotton in 1934 None

38. Give the name of any member of your family who owns land that will probably be purchased by TVA None

39. Farm expenditures and receipts for 1934:

Expenditures	Amount	Receipts	Amount
Livestock and poultry		Cotton	700
Feeds	18	Corn	
Farm implements and machinery	10	Hay	25
Machinery repairs	10	Seed	5
Farm labor	120	Livestock	
Commercial fertilizers, manure	65	Poultry and poultry products	25
Seeds, plants, trees, etc.	10	Milk and milk products	40
Taxes	10	Forest products	5
Insurance of buildings, crops	10	Other products	
Cash rent		Cons. occupation, handicrafts	200
Other (fence repair, etc.)	65	A.I.A.	
	380	Other receipts	
Total expenditures, 1934		Total receipts, 1934	1255

40. Information concerning cash income for twelve months ending Dec. 31, 1934:

Source	Amount	Source	Amount
Wages		Children away from home	
Pensions		Children at home	
Relief		Industrial compensation	1225
Insurance		Farm receipts	
Investments		Other sources	
Gifts		Total cash income	1225

41. If on relief in 1934, give estimated value of groceries, clothing, fuel, etc. (not included in cash reported) None Has any relief been received during 1935? No If so, approximate amount None

42. Family living: (Products furnished by the farm and used by the family, 1934)

Products	Quantity	Value	Products	Quantity	Value
Butter, pounds	250	25.00	Peaches, bushels	20	25.00
Milk, gallons	250	25.00	Garden, Acres	25	25.00
Eggs, dozen	200	20.00	Irish potatoes (other than in garden), bushels	25	25.00
Poultry, Number	100	25.00	Sweet potatoes (other than in garden), bushels	25	25.00
Pork, lbs. dressed	200	20.00	Honey, pounds	20	20.00
Corn, bushels	200	20.00	Wood, cords	20	20.00
Wheat, bushels	200	20.00	Other products		
Sorghum, gallons	200	20.00	Total		497.00
Peas & beans, bushels	200	20.00			
Apples, bushels	200	20.00			

43. Farm machinery:

Article	Age	Value	Article	Age	Value	Article	Age	Value
Wagon	25	15	Cultivator	15	15	Fertilizer distributor		
Wheeler	15	15	Turning plow			Small tools		
Hayrack			Grain plow	10	10	Harness		
Binder			Planters			Other		
Hay baler			Grain drills					
Stalk cutter	10	10	Tractor					
Disc harrow	10	10	Section harrow	10	10	Total Value		115

44. Has new location been definitely selected? No If so, in what county? None P. O. address None

45. Reasons for selection See capital \$83.00
Did a real estate agency assist in selection? None Some other organization None

46. If no location has been selected, does the individual desire to relocate:
a. Near his present community No
b. In the same county but another vicinity No
c. Elsewhere in the Tennessee Valley No
d. In the South but outside the Tennessee Valley No
e. Elsewhere (name) None

47. Will family buy or rent? None Approximate amount to invest None

48. Size of house desired None Is electricity desired? None

49. Acres wanted: Crop land None Pasture None Woodland None Total None

50. Has employment been secured? No If so, what type? Income 1772.00

51. What type of employment is desired: Farm None Part-time farm and industry None
Industry None Other mercantile or professional employment None

52. Special trades in which husband is skilled, such as automobile mechanic, carpenter, brick mason, etc. None

53. Give names and addresses of close relatives of this family living in this section, and give relationship to head of the family:

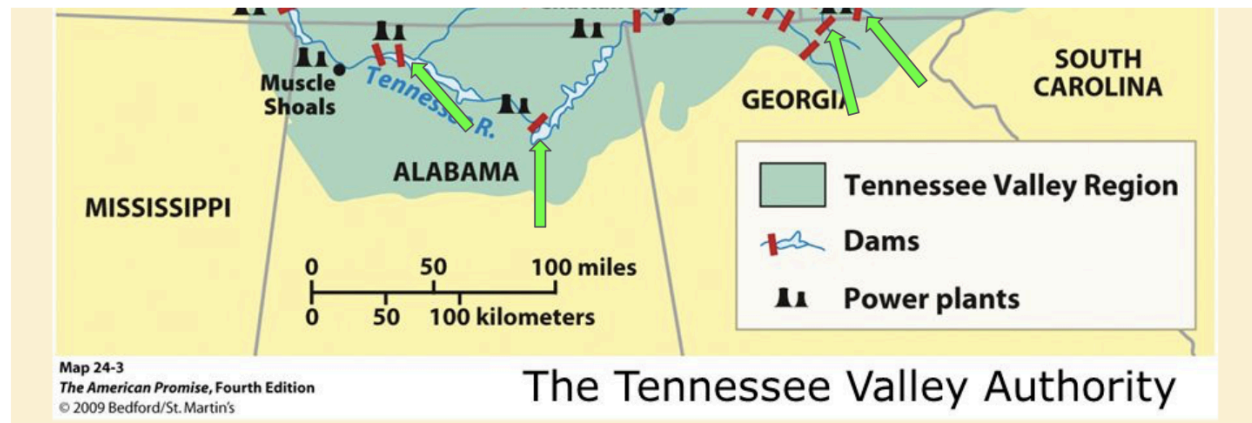
Name	Address	Relationship

54. How did the individual respond to the interview:
Antagonistic None Suspicious None Indifferent None Interested None Gladly Cooperated None

55. What is his attitude toward the TVA:
Antagonistic None Critical None Neutral None Interested None Active Resister None

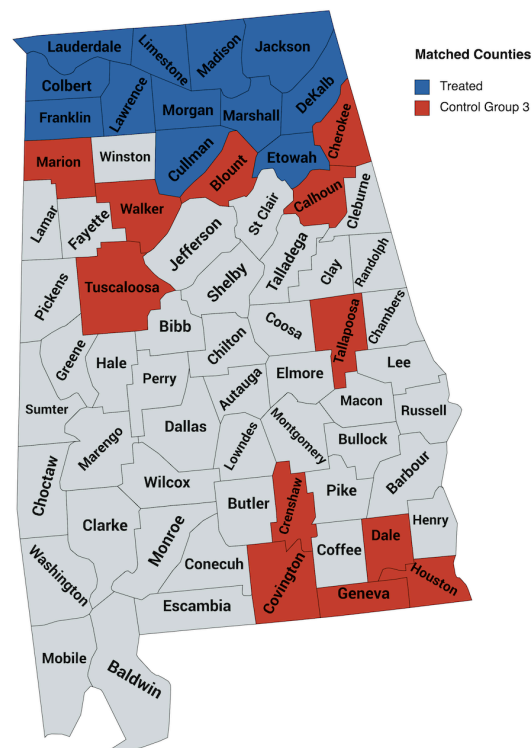
56. Write any comments regarding special problems or characteristics of this family on the back of last sheet.

Figure 3.1 Treatment and control areas



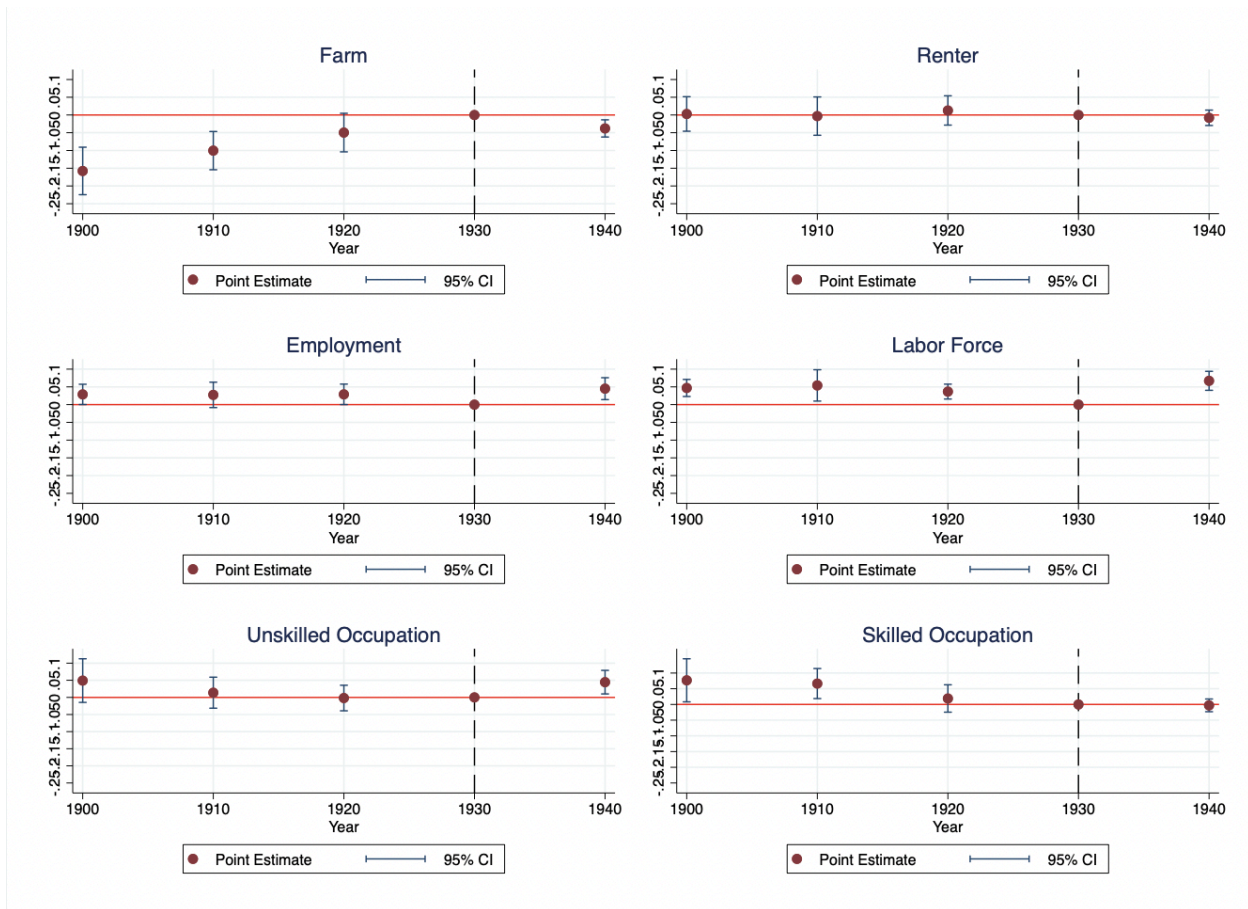
Note: Arrows point to the dams that are displacing people within our sample's time frame. Wheeler and Guntersville dams are located in Alabama, and Nottely and Chatuge dams are in Georgia. The placement of the Alabama dams displaced people in our treatment group whereas the placement of the Georgia dams displaced people in our control group in our robustness check.

Figure 3.2. Treatment and control areas



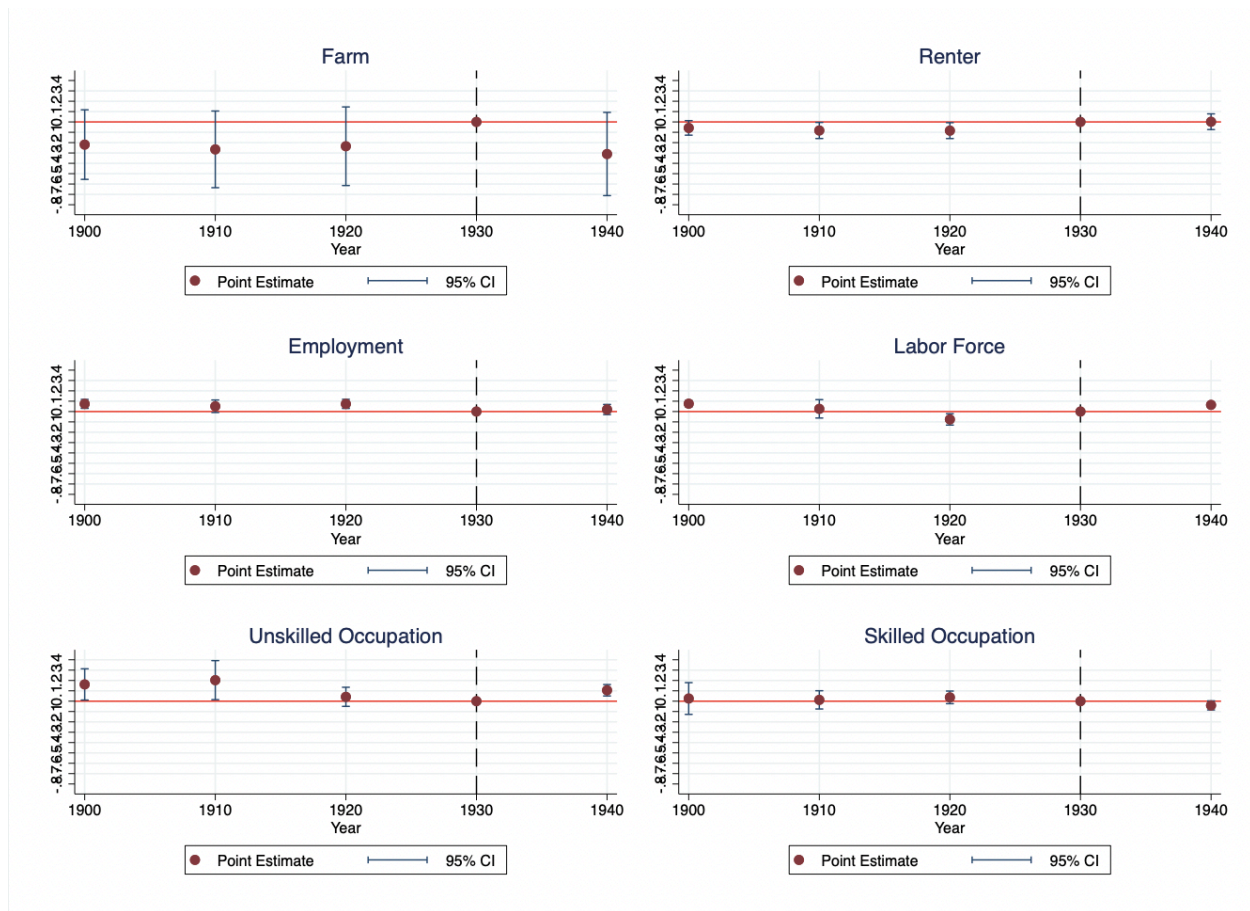
Note: Our third comparison group consists of people living in counties that are similar to those of the treated group based on 1930 county characteristics. Blue counties contain people who were displaced by the dam construction and red counties contain people who were not displaced by the dam construction,

Figure 4. Event study estimates of socioeconomic outcomes, within county control group



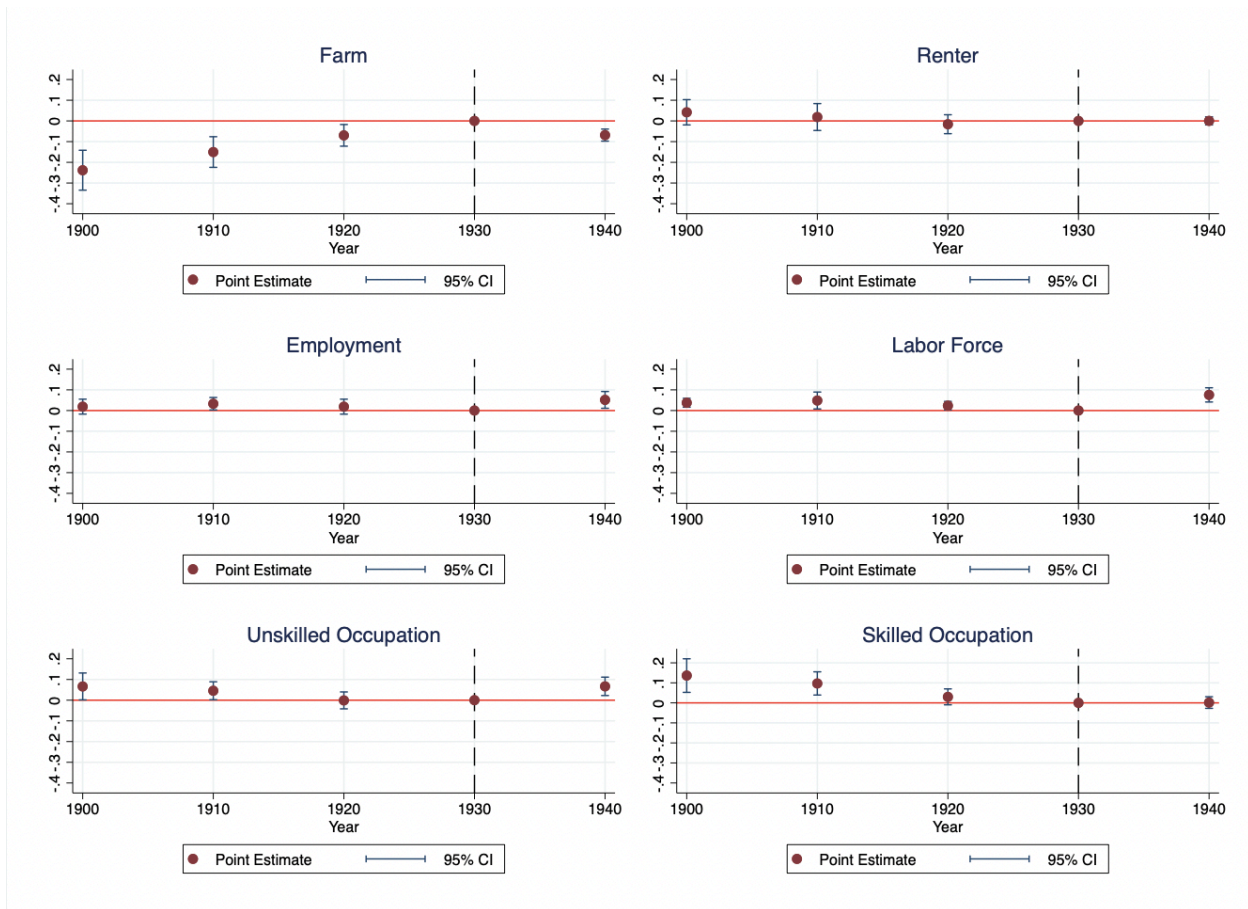
Notes: Control group includes individuals who were living in the same county of those forced to relocate, but were not in the path of the dam construction. Estimates for farm and rent are at the household level, and estimates for other outcomes are at the individual level.

Figure 5.1. Event study estimates of socioeconomic outcomes, Georgia control group



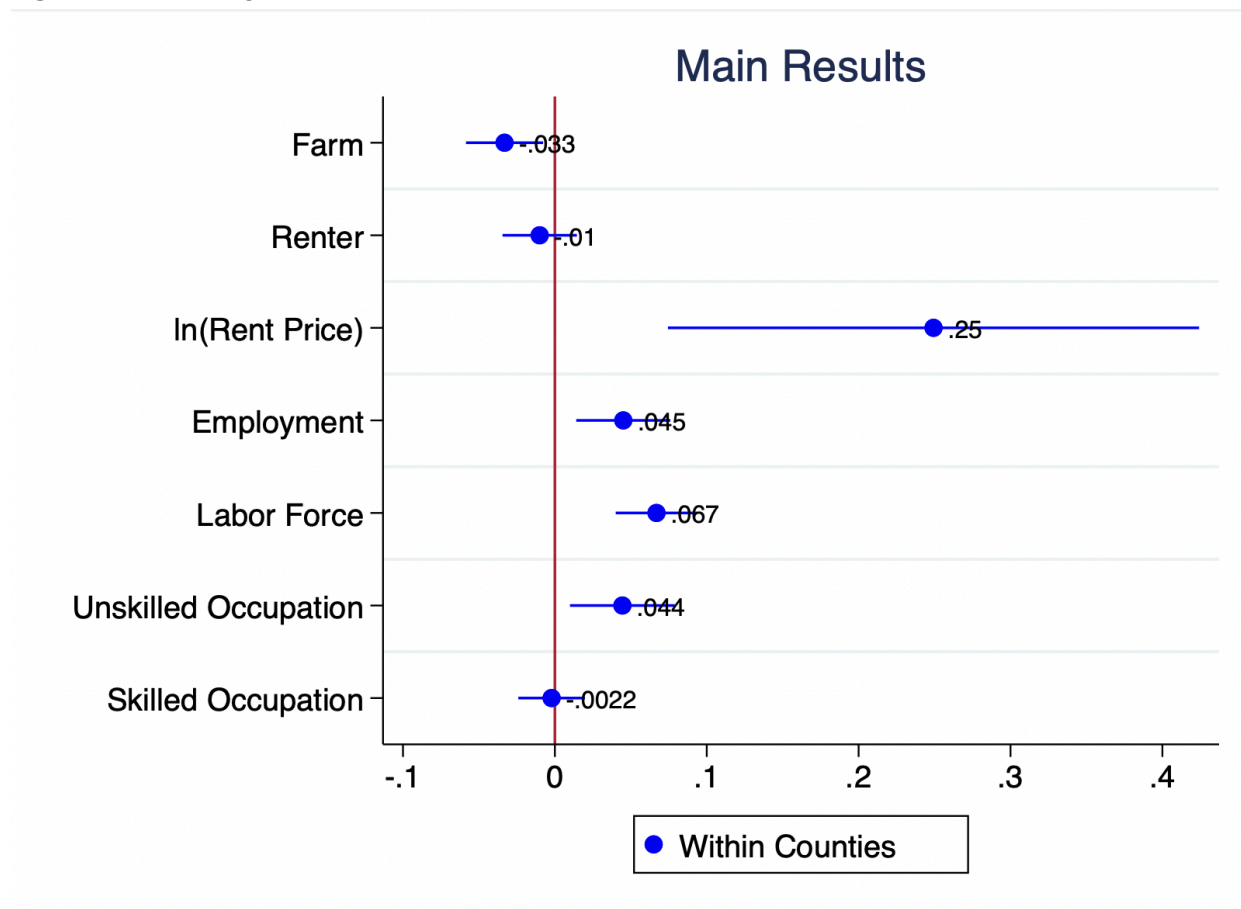
Notes: Control group includes individuals living in Georgia who have-not-yet been treated (relocated) due to a later dam construction year. Estimates for farm and rent are at the household level, and estimates for other outcomes are at the individual level.

Figure 5.2. Event study estimates of socioeconomic outcomes, matched county control group



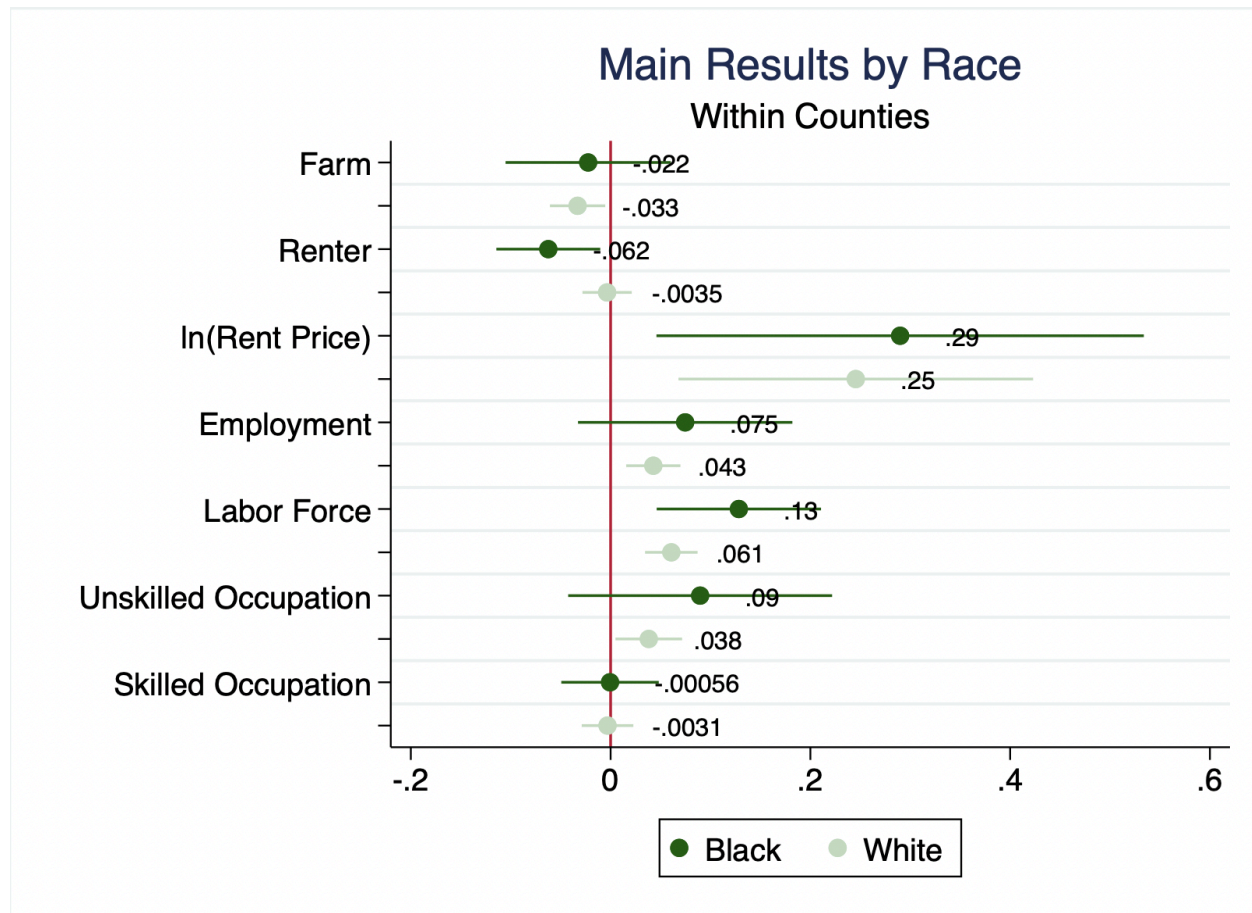
Notes: Control group includes individuals living in counties that were matched to treatment counties using a nearest neighbor matching algorithm based on 1930 county characteristics. Estimates for farm and rent are at the household level, and estimates for other outcomes are at the individual level.

Figure 6. The average treatment effect of a forced relocation, main results.



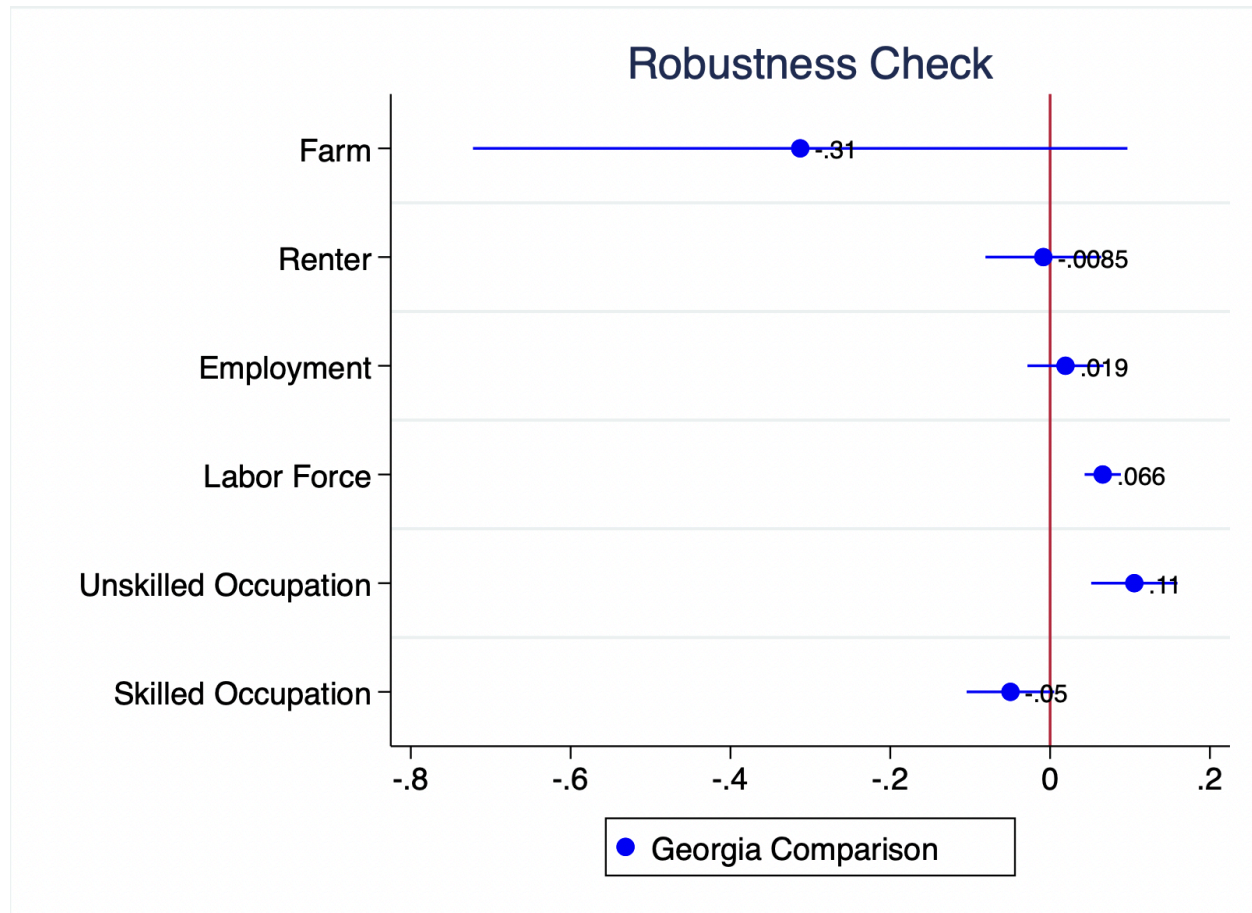
Notes: Regression coefficients by outcomes are shown graphically for our within-counties control group: individuals who were living in the same county of those forced to relocate, but were not in the path of the dam construction. The graphed bars are the standard errors.

Figure 7. The average treatment effect of a forced relocation on outcomes, main results by race



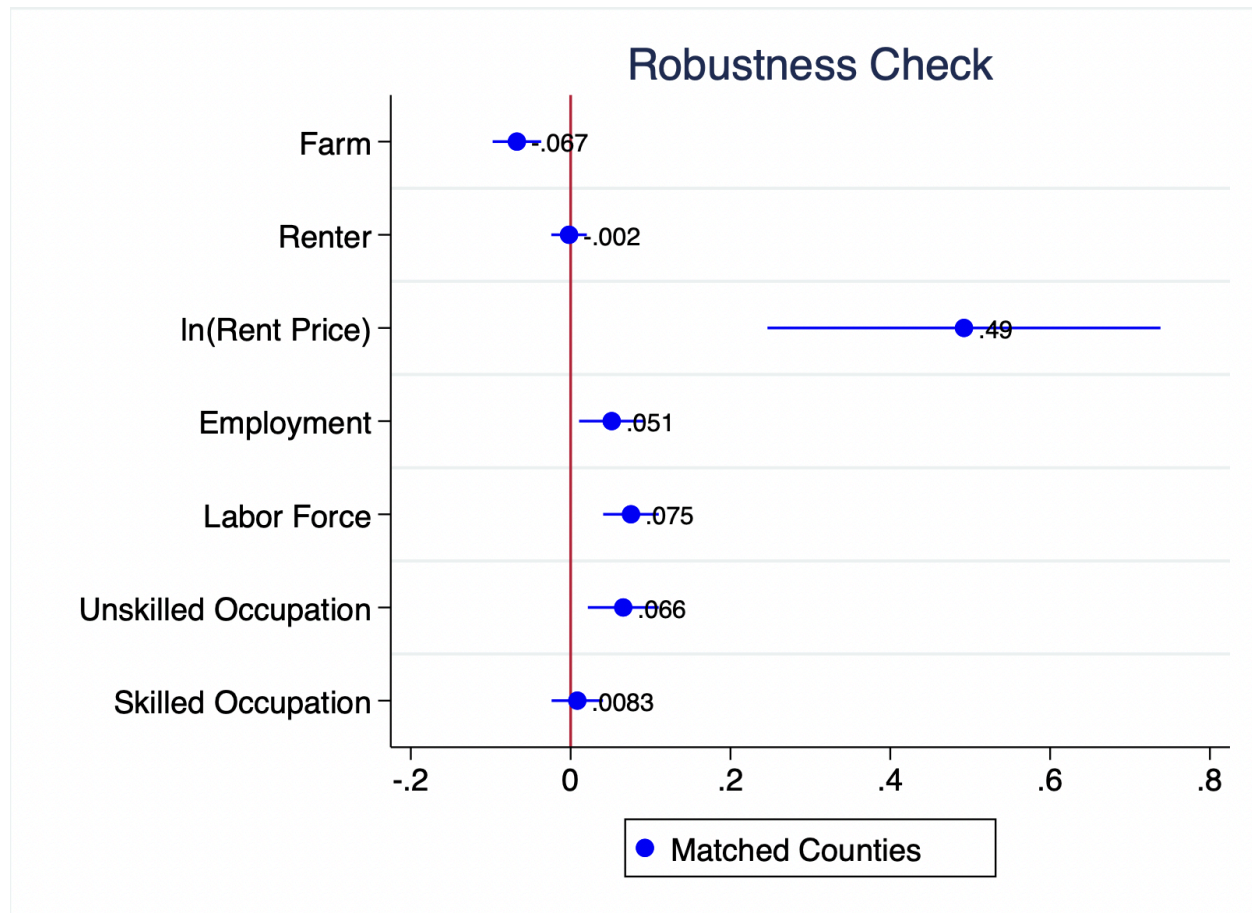
Notes: Regression coefficients by outcomes are shown graphically for our within-counties control group by race: individuals who were living in the same county of those forced to relocate, but were not in the path of the dam construction. The graphed bars are the standard errors.

Figure 8.1. The average treatment effect of a forced relocation on outcomes, robustness check - Georgia comparison



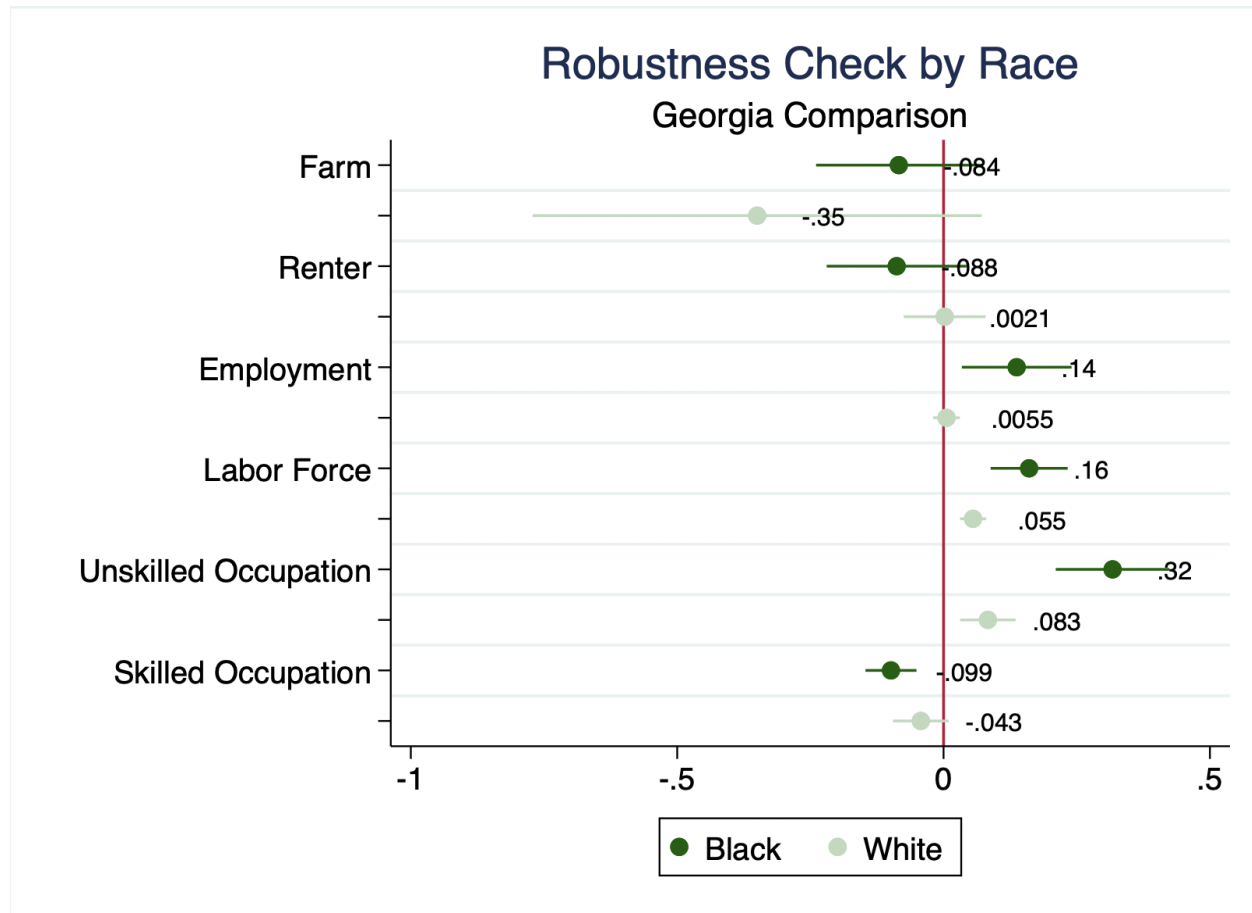
Notes: Regression coefficients by outcomes are shown graphically for our robustness check, Georgia control group: individuals living in Georgia who have-not-yet been treated (relocated) due to a later dam construction year. The graphed bars are the standard errors.

Figure 8.2 The average treatment effect of a forced relocation on outcomes, robustness check - matched counties comparison



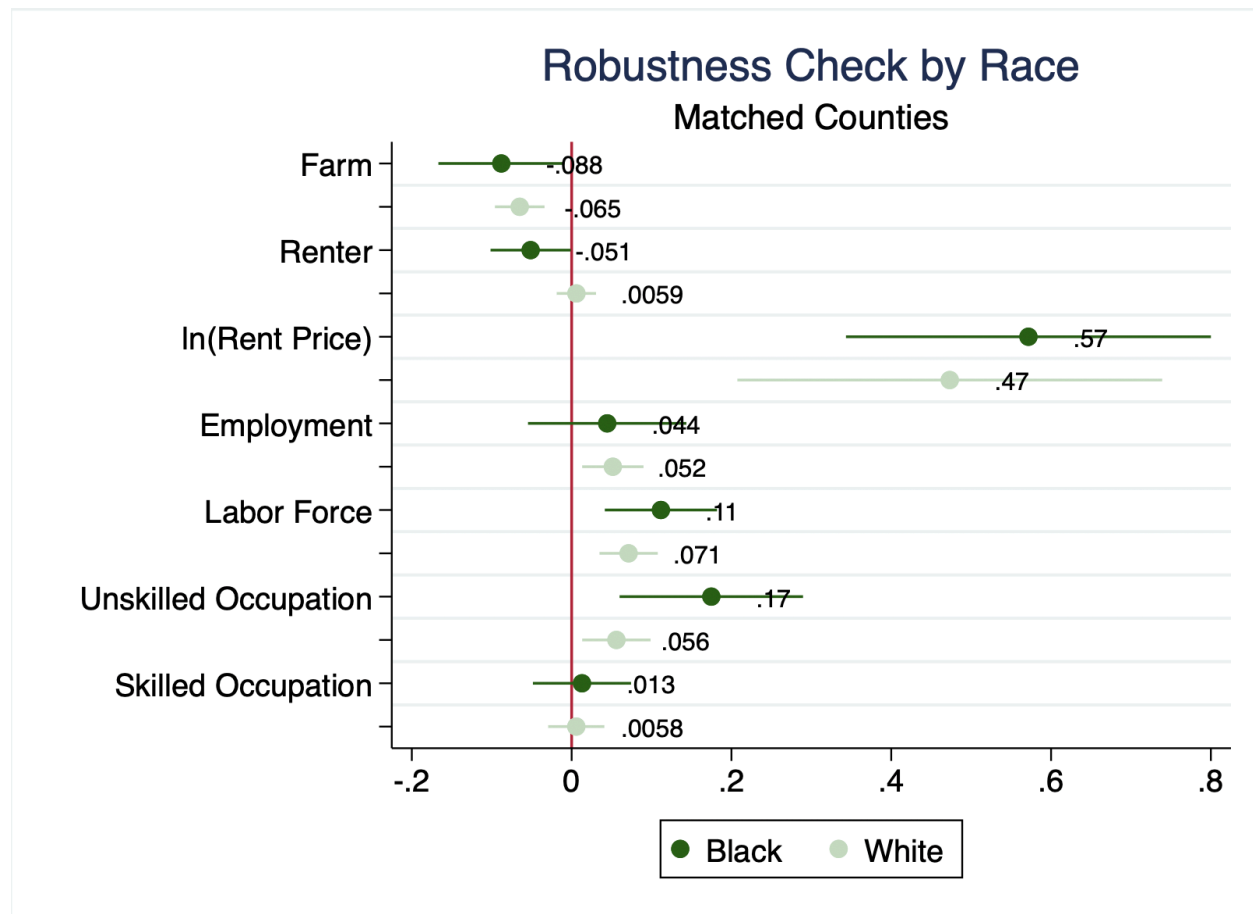
Notes: Regression coefficients by outcomes are shown graphically for our robustness check - matched counties control group: includes individuals living in counties that were matched to treatment counties using a nearest neighbor matching algorithm based on 1930 county characteristics. The graphed bars are the standard errors.

Figure 9.1. The average treatment effect of a forced relocation on outcomes by race, robustness check - Georgia comparison



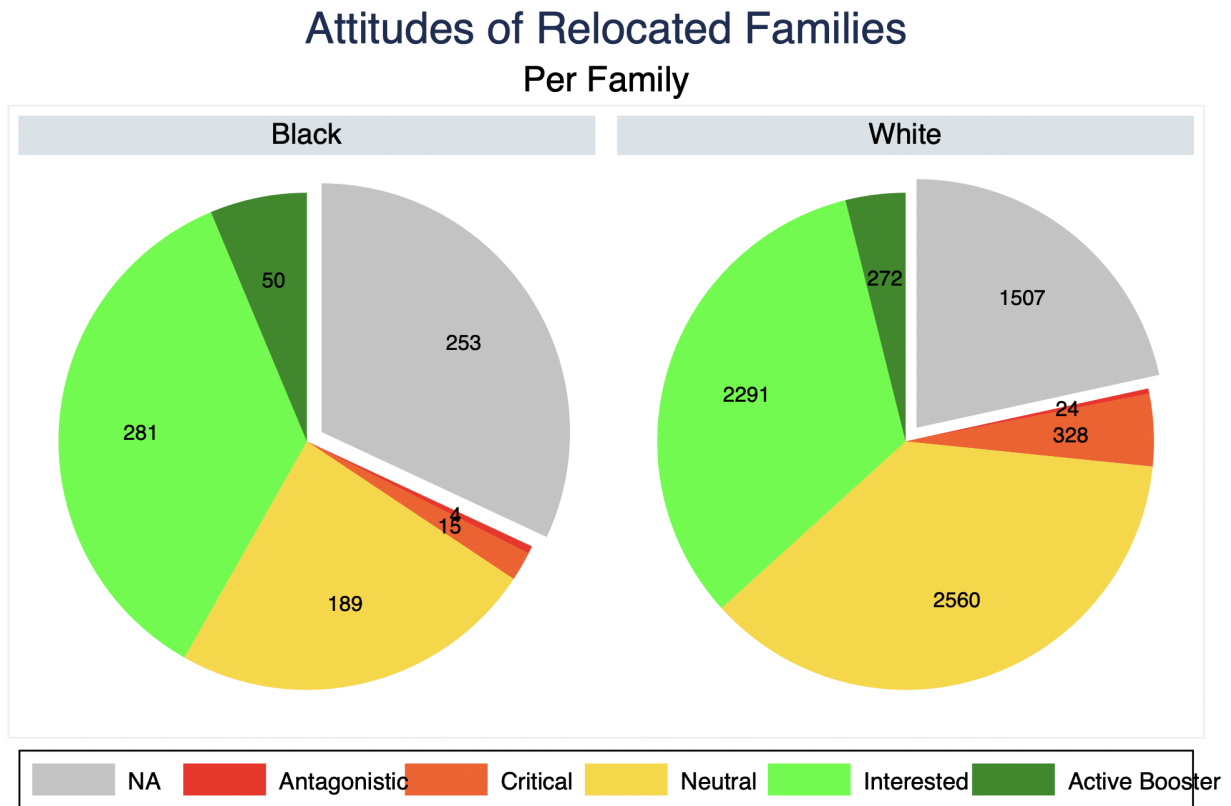
Notes: Regression coefficients by outcomes are shown graphically by race for our robustness check - Georgia control group: individuals living in Georgia who have-not-yet been treated (relocated) due to a later dam construction year. The graphed bars are the standard errors.

Figure 9.2. The average treatment effect of a forced relocation on outcomes by race, robustness check - matched counties



Notes: Regression coefficients by outcomes are shown graphically by race for our robustness check - matched counties control group: includes individuals living in counties that were matched to treatment counties using a nearest neighbor matching algorithm based on 1930 county characteristics. The graphed bars are the standard errors.

Figure 10. Attitude toward relocation for Black and White families



Notes: Data are from a survey of relocated families conducted by the TVA.

Table 1. Summary statistics for treatment and control group, on individual and household data, 1930 Census

	TG	CG	Difference
A. Individual Data			
<i>Dependent Variables</i>			
Employment	0.743 [0.437]	0.749 [0.434]	-0.006 (0.009)
Labor Force	0.704 [0.457]	0.759 [0.428]	-0.055*** (0.009)
Unskilled Jobs	0.297 [0.457]	0.279 [0.448]	0.018* (0.011)
Skilled Jobs	0.105 [0.306]	0.418 [0.493]	-0.313*** (0.007)
<i>Independent Variables</i>			
White	0.9 [0.3]	0.851 [0.356]	0.049*** (0.006)
Black	0.1 [0.3]	0.149 [0.356]	-0.049*** (0.006)
Observations	2,420	129,615	
B. Household Data			
<i>Dependent Variables</i>			
Farm	0.817 [0.386]	0.407 [0.491]	0.410*** (0.008)
Owned	1.641 [0.483]	1.575 [0.504]	0.066*** (0.011)
House Value	228.612 [1279.335]	1207.273 [4912.377]	-978.661*** (31.541)
Rent (\$/mo.)	1.535 [5.296]	9.425 [12.62]	-7.890*** (0.152)
<i>Independent Variables</i>			
White	0.906 [0.293]	0.861 [0.346]	0.045*** (0.005)
Black	0.094 [0.293]	0.139 [0.346]	-0.045*** (0.005)
Observations	2,160	107,523	

Notes: TG indicates treatment group, CG includes individuals who were living in the same county of those forced to relocate, but were not in the path of the dam construction. Data are from the 1930 US Census. Standard deviations in brackets, standard errors in parentheses. *** p<0.01

Table 2. Effect of TVA relocation on socioeconomic outcomes

VARIABLES	(1) Farm	(2) Renter	(3) ln(Rent \$)	(4) Employment	(5) Labor Force	(6) Unskilled Occupation	(7) Skilled Occupation
<u>A. Within Counties Comparison</u>							
Treat*Post	-0.035** (0.013)	-0.007 (0.011)	0.253*** (0.081)	0.045*** (0.014)	0.067*** (0.012)	0.044** (0.016)	-0.002 (0.010)
Observations	120,804	120,804	66,028	148,432	148,432	124,649	124,649
R-squared	0.072	0.021	0.284	0.014	0.041	0.023	0.074
Mean	0.674	0.586	0.712	0.635	0.612	0.323	0.210
<u>B. Georgia Have-Not-Yet Been Treated</u>							
Treat*Post	-0.311 (0.193)	-0.005 (0.034)	1.499*** (0.070)	0.019 (0.022)	0.066*** (0.011)	0.105*** (0.025)	-0.050* (0.026)
Observations	6,460	6,460	3,391	5,089	5,089	4,267	4,267
R-squared	0.040	0.034	0.364	0.030	0.075	0.020	0.039
Mean	0.788	0.596	0.325	0.535	0.504	0.340	0.118
<u>C. Matched Counties</u>							
Treat*Post	-0.073*** (0.014)	-0.002 (0.011)	0.489*** (0.121)	0.051** (0.020)	0.075*** (0.017)	0.066*** (0.021)	0.008 (0.016)
Observations	106,191	106,191	60,440	127,381	127,381	105,053	105,053
R-squared	0.039	0.024	0.206	0.012	0.035	0.033	0.064
Mean	0.575	0.614	0.862	0.602	0.579	0.353	0.272

Notes: Main effects for treatment and year fixed effects are included in each regression. Race controls are included in each regression. Control group 1 includes individuals who were living in the same county of those forced to relocate, but were not in the path of the dam construction, control group 2 indicates individuals living in Georgia who have-not-yet been treated (relocated) due to a later dam construction year, and control group 3 includes individuals living in counties that were matched to treatment counties using a nearest neighbor matching algorithm based on 1930 county characteristics. Robust standard errors in parentheses, *** p<0.01, ** p<0.05, * p<0.1

Table 3. Effect of TVA relocation by race, within-counties control group

VARIABLES	(1) Farm	(2) Renter	(3) ln(Rent \$)	(4) Employment	(5) Labor Force	(6) Unskilled Occupation	(7) Skilled Occupation
<u>A. White</u>							
Treat*Post	-0.036** (0.013)	-0.002 (0.011)	0.246** (0.083)	0.043*** (0.013)	0.061*** (0.012)	0.038** (0.015)	-0.003 (0.012)
Observations	109,723	109,723	58,385	135,556	135,556	113,723	113,723
R-squared	0.071	0.014	0.276	0.016	0.042	0.003	0.066
Mean	0.677	0.573	0.739	0.626	0.603	0.308	0.224
<u>B. Black</u>							
Treat*Post	-0.018 (0.042)	-0.051** (0.020)	0.310** (0.111)	0.075 (0.050)	0.128*** (0.038)	0.090 (0.061)	-0.001 (0.023)
Observations	11,064	11,064	7,633	12,853	12,853	10,904	10,904
R-squared	0.143	0.010	0.359	0.004	0.026	0.022	0.053
Mean	0.640	0.722	0.496	0.751	0.724	0.481	0.0715
Coefficients Different by Race	No	Yes	No	No	Yes	No	No

Notes: Main effects for treatment and year fixed effects are included in each regression. Control group includes individuals who were living in the same county of those forced to relocate, but were not in the path of the dam construction. Robust standard errors in parentheses. *** p<0.01, ** p<0.05, * p<0.1

Table 4. Effect of TVA relocation by race, Georgia control group

VARIABLES	(1) Farm	(2) Renter	(3) ln(Rent \$)	(4) Employment	(5) Labor Force	(6) Unskilled Occupation	(7) Skilled Occupation
<u>A. White</u>							
Treat*Post	-0.351* (0.197)	0.006 (0.036)	1.511*** (0.077)	0.005 (0.012)	0.055*** (0.012)	0.083*** (0.025)	-0.043 (0.025)
Observations	5,806	5,806	2,997	4,561	4,561	3,822	3,822
R-squared	0.044	0.024	0.357	0.032	0.074	0.002	0.030
Mean	0.788	0.575	0.345	0.524	0.497	0.320	0.129
<u>B. Black</u>							
Treat*Post	-0.068 (0.072)	-0.090 (0.060)	1.401*** (0.084)	0.137** (0.048)	0.161*** (0.033)	0.317*** (0.049)	-0.099*** (0.022)
Observations	648	648	391	520	520	438	438
R-squared	0.068	0.041	0.419	0.029	0.092	0.043	0.069
Mean	0.791	0.782	0.177	0.629	0.571	0.490	0.0307
Coefficients	No	No	No	Yes	Yes	Yes	No
Different by Race							

Notes: Main effects for treatment and year fixed effects are included in each regression. Control group includes individuals living in Georgia who have-not-yet been treated (relocated) due to a later dam construction year. Robust standard errors in parentheses, *** p<0.01, ** p<0.05, * p<0.1

Table 5. Effect of TVA relocation by race, matched county controls

VARIABLES	(1) Farm	(2) Renter	(3) ln(Rent \$)	(4) Employment	(5) Labor Force	(6) Unskilled Occupation	(7) Skilled Occupation
<u>A. White</u>							
Treat*Post	-0.070*** (0.015)	0.005 (0.012)	0.470*** (0.131)	0.052** (0.019)	0.071*** (0.018)	0.056** (0.021)	0.006 (0.017)
Observations	91,587	91,587	50,130	110,944	110,944	91,134	91,134
R-squared	0.034	0.011	0.195	0.013	0.035	0.008	0.046
Mean	0.580	0.593	0.900	0.589	0.568	0.328	0.294
<u>B. Black</u>							
Treat*Post	-0.098** (0.036)	-0.040* (0.022)	0.569*** (0.113)	0.044 (0.048)	0.112*** (0.034)	0.175*** (0.056)	0.013 (0.030)
Observations	14,580	14,580	10,300	16,409	16,409	13,892	13,892
R-squared	0.129	0.019	0.259	0.009	0.032	0.010	0.088
Mean	0.542	0.747	0.668	0.699	0.669	0.514	0.127
Coefficients Different by Race	No	Yes	No	No	No	Yes	No

Notes: Main effects for treatment and year fixed effects are included in each regression. Control group includes individuals living in counties that were matched to treatment counties using a nearest neighbor matching algorithm based on 1930 county characteristics. Robust standard errors in parentheses,

*** p<0.01, ** p<0.05, * p<0.1

Table 6. Effect of TVA relocation on labor market outcomes, by attitude

VARIABLES	<u>Control Group 1</u>			<u>Control Group 2</u>		
	Negative	Neutral	Positive	Negative	Neutral	Positive
<u>A. Labor Force</u>						
Treat*Post	0.006 (0.069)	0.060*** (0.009)	0.038*** (0.013)	-0.009 (0.069)	0.044*** (0.008)	0.022* (0.012)
<i>1930 Mean</i>	<i>0.733</i>	<i>0.732</i>	<i>0.732</i>	<i>0.728</i>	<i>0.728</i>	<i>0.728</i>
Observations	192,726	195,400	195,666	213,307	215,981	216,247
R-squared	0.277	0.278	0.277	0.287	0.287	0.287
<u>B. Unskilled Jobs</u>						
Treat*Post	0.031 (0.039)	0.038** (0.017)	0.069*** (0.019)	0.039 (0.038)	0.046*** (0.014)	0.077*** (0.018)
<i>1930 Mean</i>	<i>0.258</i>	<i>0.258</i>	<i>0.257</i>	<i>0.241</i>	<i>0.241</i>	<i>0.240</i>
Observations	192,726	195,400	195,666	213,307	215,981	216,247
R-squared	0.016	0.016	0.016	0.018	0.018	0.018
<u>B. Skilled Jobs</u>						
Treat*Post	-0.076*** (0.025)	-0.055** (0.019)	-0.057*** (0.018)	-0.109*** (0.028)	-0.086*** (0.021)	-0.089*** (0.022)
<i>1930 Mean</i>	<i>0.080</i>	<i>0.061</i>	<i>0.071</i>	<i>0.080</i>	<i>0.061</i>	<i>0.071</i>
Observations	192,726	195,400	195,666	213,307	215,981	216,247
R-squared	0.088	0.089	0.088	0.108	0.109	0.109
<u>C. Employment</u>						
Treat*Post	0.039 (0.036)	0.059*** (0.008)	0.033** (0.014)	0.020 (0.037)	0.040*** (0.010)	0.014 (0.016)
<i>1930 Mean</i>	<i>0.769</i>	<i>0.768</i>	<i>0.768</i>	<i>0.757</i>	<i>0.756</i>	<i>0.756</i>
Observations	192,726	195,400	195,666	213,307	215,981	216,247
R-squared	0.526	0.526	0.526	0.528	0.529	0.528

Notes: Main effects for treatment and year fixed effects are included in each regression. Race controls are included in each regression. Negative includes individuals who were antagonistic or critical of the relocation program. Positive includes individuals who were interested or active boosters of the relocation program. Control group 1 includes individuals who were living downriver of the dam, and control group 2 indicates individuals living in counties parallel to the counties bordering the Tennessee River. Robust standard errors in parentheses, *** p<0.01, ** p<0.05, * p<0.1

Appendix

Figure A1. Stages of dam construction mentioned by Howard (1936)

Step 1: Both engineers and social scientists must make numerous investigations. The engineers want to find a firm, physical foundation for their structure and for the finished product to withstand the forces of nature. Before the project is approved or the site is determined, the social scientists need to figure out if the anticipated benefits outweigh the monetary and social costs of the enterprise. If so, they should proceed with the dam construction.

Step 2: After the decision to build the dam has been made, there should be focus on questions relating to construction. What labor supply is available within a reasonable distance of the project? What is the composition of the population? What skills are represented by the population? What race factors are present among the population?

Step 3: Consideration of the need for a camp to house employees after they are hired which includes awareness of all social implications involved in housing workers in construction communities.

Step 4: Understanding the people who are to be employed. What skills are needed? What type of employee training will be most useful? Will the training be on the construction site or will it be held at workers homes?

Step 5: Studies are made to assist in planning the educational facilities needed by the employees of the Authority.

Step 6: As the actual construction begins, there are a new series of questions to be considered. What commercial facilities must be provided? Are there any desirable recreational facilities? What other economic and social needs must be anticipated?

Step 7: Determine the type of community organization that will be most effective such as the community making decisions within popular participation.

Step 8: Figure out what cooperative arrangement can be worked out that will be mutually satisfactory to the Authority and the local governments. An example is to consider a reexamination of health facilities due to new demands.

Step 9: Social scientists must determine detailed facts concerning families to be displaced by the reservoir.

Step 10: As reservoir land is acquired and the people move out of the region, more questions for the public administration and economist. What will be the financial status of the local government after a large portion of an area has been purchased? What source of revenue is left? What is the debt and can it be paid? What is the future of this particular part of the local government?

Step 11: As the construction project nears completion, new problems arise. What vocational opportunities are open to workers whose period of employment is about to end? Can the Authority assist them in their efforts to utilize the experience and training received during this period of employment?

Step 12: Finally, important adjustments must be made after the construction project is over and people who were involved begin to leave the region.

Table A1. Difference-in-differences on average county characteristics, based on 1940 location, individual data

	Difference 1	Difference 2	Difference 3
Population	8471.865 (7742.19) <i>200373.47</i>	8732.001*** (1348.281) <i>38904.047</i>	-6801.373 (4044.068) <i>42357.79</i>
Male	0.001 (0.001) <i>0.499</i>	-0.002*** (0.000) <i>0.504</i>	0.001 (0.001) <i>0.501</i>
White	-0.004 (0.01) <i>0.764</i>	0.004 (0.011) <i>0.878</i>	0.012 (0.012) <i>0.786</i>
Black	0.003 (0.009) <i>0.236</i>	-0.005 (0.01) <i>0.122</i>	-0.013 (0.012) <i>0.214</i>
Hispanic	0.000 (0.000) <i>0.001</i>	0.001*** (0.000) <i>0.000</i>	0.000 (0.000) <i>0.000</i>
Age	-0.21 (0.183) <i>24.936</i>	1.878*** (0.205) <i>23.952</i>	-0.206 (0.215) <i>24.307</i>
In School	-0.001 (0.004) <i>0.24</i>	-0.008* (0.004) <i>0.247</i>	0.002 (0.005) <i>0.248</i>
Labor Force	0.001 (0.005) <i>0.357</i>	-0.003 (0.005) <i>0.323</i>	-0.006 (0.008) <i>0.339</i>
Employment	-0.003 (0.009) <i>0.356</i>	-0.035*** (0.01) <i>0.336</i>	0.000 (0.014) <i>0.356</i>
Occupation Score	0.031** (0.012) <i>0.145</i>	0.059*** (0.014) <i>0.133</i>	0.587 (0.998) <i>28.689</i>
Earnings Score	0.575 (0.591) <i>32.209</i>	-1.023** (0.436) <i>30.695</i>	-0.105 (0.578) <i>31.047</i>
Education Score	0.272 (0.444) <i>31.88</i>	-1.245*** (0.361) <i>31.339</i>	-0.353 (0.489) <i>30.878</i>

Notes: Data are average county level characteristics from the 1930 census. Differences are from 1930 and 1940 between treated and control groups 1, 2, and 3 respectively, based on where an individual was living in each census year. Standard errors are in parentheses. Italicized are the 1930 means of both treatment group and respective control group. *** p<0.01, ** p<0.05, * p<0.1

Table A2. Difference-in-differences on average county characteristics, based on 1940 location, household data

	Difference 1	Difference 2	Difference 3
SEA SD	-0.497** (0.230) <i>17.415</i>	1.683*** (0.302) <i>9.277</i>	1.910** (0.774) <i>45.078</i>
Renter	-0.002 (0.006) <i>0.628</i>	0.013 (0.008) <i>0.615</i>	0.003 (0.011) <i>0.649</i>
Farmer	-0.077** (0.035) <i>0.380</i>	-0.095*** (0.009) <i>0.708</i>	-0.052*** (0.016) <i>0.513</i>
Rural	-0.065** (0.028) <i>0.572</i>	-0.064*** (0.012) <i>0.868</i>	-0.013 (0.017) <i>0.783</i>

Notes: Data are average county level characteristics from the 1930 census. Differences are from 1930 and 1940 between treated and control groups 1, 2, and 3 respectively, based on where an individual was living in each census year. SEA stands for State Economic Area which represents groups of counties that have similar economic characteristics when initially defined. Standard errors are in parentheses. Italicized are the 1930 means of both treatment group and respective control group.

*** p<0.01, ** p<0.05, * p<0.1

Table A3. Difference-in-differences on average county characteristics, based on 1940 location, individual by race

	Black			White		
	Difference 1	Difference 2	Difference 3	Difference 1	Difference 2	Difference 3
Population	9591.032 (5763.679) <i>308245.09</i>	7731.761** (3038.551) <i>37993.715</i>	-11861.102** (5003.828) <i>44,444.82</i>	8341.900 (8257.368) <i>183131.39</i>	8867.474*** (1617.034) <i>39015.543</i>	-6062.853 (4195.355) <i>42,049.79</i>
Male	0.003*** (0.001) <i>0.495</i>	-0.002** (0.001) <i>0.501</i>	0.000 (0.001) <i>0.498</i>	0.001 (0.001) <i>0.499</i>	-0.003*** (0.000) <i>0.504</i>	0.001 (0.001) <i>0.501</i>
Age	-0.182 (0.107) <i>25.7</i>	1.969*** (0.102) <i>23.937</i>	-0.120 (0.134) <i>24.685</i>	-0.209 (0.192) <i>24.814</i>	1.867*** (0.218) <i>23.954</i>	-0.215 (0.227) <i>24.252</i>
In School	-0.002 (0.004) <i>0.227</i>	-0.003 (0.006) <i>0.244</i>	0.009 (0.007) <i>0.245</i>	0.000 (0.005) <i>0.242</i>	-0.009** (0.004) <i>0.247</i>	0.002 (0.005) <i>0.249</i>
Labor Force	-0.004 (0.008) <i>0.381</i>	-0.010 (0.01) <i>0.329</i>	-0.021 (0.011) <i>0.347</i>	0.002 (0.004) <i>0.353</i>	-0.002 (0.004) <i>0.322</i>	-0.004 (0.007) <i>0.334</i>
Employment	-0.022 (0.015) <i>0.365</i>	-0.049** (0.018) <i>0.347</i>	-0.027 (0.018) <i>0.360</i>	-0.001 (0.009) <i>0.354</i>	-0.034*** (0.009) <i>0.335</i>	0.004 (0.013) <i>0.355</i>
Occupation Score	0.014 (0.013) <i>0.150</i>	0.041** (0.014) <i>0.155</i>	-1.326 (1.131) <i>29.107</i>	0.033** (0.012) <i>0.144</i>	0.061*** (0.014) <i>0.131</i>	0.830 (0.987) <i>28.627</i>
Earnings Score	1.657** (0.766) <i>33.076</i>	-0.379 (0.726) <i>30.205</i>	0.897 (0.742) <i>31.120</i>	0.428 (0.555) <i>32.07</i>	-1.093** (0.391) <i>30.755</i>	-0.229 (0.565) <i>31.036</i>
Education Score	1.118* (0.592) <i>32.224</i>	-0.710 (0.591) <i>30.835</i>	0.539 (0.612) <i>30.877</i>	0.157 (0.419) <i>31.825</i>	-1.305*** (0.325) <i>31.401</i>	-0.467 (0.482) <i>30.878</i>

Notes: Data are average county level characteristics from the 1930 census. Differences are from 1930 and 1940 between treated and control groups 1, 2, and 3 respectively, based on where an individual was living in each census year. Standard errors are in parentheses. Italicized are the 1930 means of both treatment group and respective control group. *** p<0.01, ** p<0.05, * p<0.1

Table A4. Difference-in-differences on average county characteristics, based on 1940 location, household by race

	Black			White		
	Difference 1	Difference 2	Difference 3	Difference 1	Difference 2	Difference 3
SEA SD	-0.093 (1.183) <i>19.756</i>	4.129*** (1.117) <i>16.55</i>	4.075** (1.728) <i>48.123</i>	-0.427 (0.267) <i>16.955</i>	1.445*** (0.340) <i>8.372</i>	1.706** (0.750) <i>44.596</i>
Renter	-0.013 (0.008) <i>0.657</i>	0.003 (0.011) <i>0.658</i>	-0.009 (0.014) <i>0.671</i>	-0.001 (0.006) <i>0.622</i>	0.014* (0.007) <i>0.610</i>	0.005 (0.011) <i>0.645</i>
Farmer	-0.112** (0.038) <i>0.212</i>	-0.106*** (0.016) <i>0.702</i>	-0.062** (0.023) <i>0.482</i>	-0.071* (0.034) <i>0.414</i>	-0.095*** (0.010) <i>0.709</i>	-0.051*** (0.016) <i>0.517</i>
Rural	-0.073** (0.026) <i>0.424</i>	-0.054*** (0.012) <i>0.873</i>	-0.040 (0.017) <i>0.743</i>	-0.063** (0.028) <i>0.601</i>	-0.065*** (0.012) <i>0.867</i>	-0.014 (0.017) <i>0.790</i>

Notes: Data are average county level characteristics from the 1930 census. Differences are from 1930 and 1940 between treated and control groups 1, 2, and 3 respectively, based on where an individual was living in each census year. SEA stands for State Economic Area which represents groups of counties that have similar economic characteristics when initially defined. Standard errors are in parentheses. Italicized are the 1930 means of both treatment group and respective control group. *** p<0.01, ** p<0.05, * p<0.1

Table A5. Effect of relocation on occupation score

	Within Counties Comparison	Georgia Comparison	Matched Counties Comparison
<u>A. All Individuals</u>			
Treat*Post	-3.288* (1.844)	1.683 (1.001)	-0.162 (1.582)
Observations	427,068	9,550	225,969
R-squared	0.2162	0.1836	0.1716
<i>Mean</i>	<i>54.186</i>	<i>51.567</i>	<i>49.735</i>
<u>B. White</u>			
Treat*Post	-3.686 (2.169)	0.990 (1.142)	-0.4250 (1.720)
Observations	362,057	8,550	196,127
R-squared	0.2178	0.1854	0.1736
<i>Mean</i>	<i>53.754</i>	<i>51.674</i>	<i>49.816</i>
<u>C. Black</u>			
Treat*Post	-2.105 (2.173)	6.745** (2.697)	0.854 (2.004)
Observations	64,967	991	29,811
R-squared	0.2120	0.1730	0.1619
<i>Mean</i>	<i>56.669</i>	<i>50.624</i>	<i>49.176</i>

Table A6. Effect of TVA relocation by race, nearest neighbor matched sample

VARIABLES	(1) Farm	(2) Renter	(3) ln(Rent \$)	(4) Employment	(5) Labor Force	(6) Unskilled Occupation	(7) Skilled Occupation
<u>A. White</u>							
Treat*Post	-0.111*** (0.026)	-0.045 (0.049)	0.148 (0.092)	0.099** (0.036)	0.141** (0.049)	0.141*** (0.025)	-0.028 (0.021)
Observations	8,790	8,790	8,790	12,073	12,073	12,073	12,073
R-squared	0.061	0.012	0.116	0.005	0.028	0.011	0.074
1930 Mean	0.717	0.611	0.375	0.795	0.771	0.358	0.0644
<u>B. Black</u>							
Treat*Post	-0.036 (0.036)	-0.040** (0.014)	0.266** (0.122)	0.096* (0.052)	0.140*** (0.041)	0.144** (0.060)	-0.012 (0.016)
Observations	9,423	9,423	9,423	11,672	11,672	11,672	11,672
R-squared	0.142	0.013	0.163	0.005	0.026	0.024	0.045
1930 Mean	0.658	0.711	0.323	0.798	0.774	0.366	0.0582
Coefficients Different by Race	Yes	No	No	No	No	No	No

Notes: Main effects for treatment and year fixed effects are included in each regression. Race controls are included in each regression. Control group consists of individuals in our within-counties control group who were living in the same county of those forced to relocate, but were not in the path of the dam construction. Robust standard errors in parentheses, *** p<0.01, ** p<0.05, * p<0.1

Table A7. Characteristics of relocated families that did and did not link to the 1930s census

Linked	(1) White	(2) Male	(3) Average Income	(4) Rented	(5) Farmers	(6) Completed 12th grade or higher	(7) Total Individuals
<i>Step 1: Linking all treated Alabama residents to the 1930s census</i>							
Yes	81.91%	57.22%	\$456.12	78.33%	66.14%	2.5%	14746
No	79.59%	43.20%	\$438.62	80.06%	69.85%	2.5%	9215
<i>Step 2: Linking the above 14746 to at least one of the prior census' (1900-1920)</i>							
Yes	85.24%	66.27%	\$526.23	72.44%	72.68%	4.9%	3652
No	80.81%	54.24%	\$434.64	80.27%	63.98%	1.7%	11094
<i>Step 3: Final comparison of relocated individuals in our sample to relocated individuals who did not link</i>							
Yes	85.24%	66.27%	\$526.23	72.44%	72.68%	4.9%	3652
No	80.26%	49.23%	\$435.37	80.17%	66.64%	2.3%	20309

Notes: Each percentage serves as a description of the sample of relocated families and individuals that did and did not link to the historical census'. We start with an initial linking of relocatees to the 1930s census, as it serves as the pre period in our Difference-in-Difference analysis. We then reduced our sample to relocatees who linked to the 1940s census (post period) and to at least one of the prior census', 1900-1920, to establish parallel trends. Step 3 shows our final linked relocated sample in comparison to all relocatees who did not link in the linkage process.

Table A8. Effect of housing and labor variables from 1930 on the probability of being in the treatment group

Variables	Control Group 1	Control Group 2	Control Group 3
<u>A. Household Level</u>			
ln(Housing Value)	0.004 (0.006)	0.000 (0.000)	0.000 (0.000)
Renter	-0.009 (0.007)	-0.000* (0.000)	-0.000 (0.001)
ln(Rent Price)	0.001 (0.001)	0.000 (0.000)	0.001 (0.001)
Farmer	0.021** (0.008)	0.000 (0.000)	0.002 (0.002)
<u>B. Individual Level</u>			
Labor force	-0.005 (0.007)	-0.000 (0.000)	-0.000 (0.001)
Employment	0.004 (0.003)	0.000* (0.000)	-0.002 (0.001)
Unskilled Occupation	-0.007** (0.002)	-0.000 (0.000)	-0.000 (0.000)
Skilled Occupation	-0.010** (0.004)	0.000 (0.000)	-0.001 (0.001)
Household Observations	61,732	1,817	33,200
Individual Observations	99,331	2,022	51,799

Notes: We regress each outcome on treatment status in the year 1930. Race controls are included in each regression. Control group 1 includes individuals who were living in the same county of those forced to relocate, but were not in the path of the dam construction, control group 2 indicates individuals living in Georgia who have-not-yet been treated (relocated) due to a later dam construction year, and control group 3 includes individuals living in counties that were matched to treatment counties using a nearest neighbor matching algorithm based on 1930 county characteristics. Robust standard errors in parentheses, *** p<0.01, ** p<0.05, * p<0.1

Table A9. Effect of TVA relocation by home ownership, within-counties control group

VARIABLES	(1) Farm	(2) Employment	(3) Labor Force	(4) Unskilled Occupation	(5) Skilled Occupation
<u>A. Renter</u>					
Treat*Post	-0.046** (0.020)	0.062*** (0.013)	0.092*** (0.015)	0.050** (0.020)	0.009 (0.013)
Observations	68,064	84,102	84,102	72,021	72,021
R-squared	0.079	0.013	0.043	0.022	0.079
Mean	0.663	0.666	0.649	0.320	0.201
<u>B. Owner</u>					
Treat*Post	0.007 (0.013)	0.023 (0.023)	0.029 (0.019)	0.011 (0.018)	-0.022 (0.013)
Observations	52,709	64,330	64,330	52,628	52,628
R-squared	0.076	0.015	0.038	0.038	0.072
Mean	0.690	0.598	0.567	0.327	0.223
Coefficients Different	No	Yes	Yes	Yes	Yes

Notes: Main effects for treatment and year fixed effects are included in each regression. Control group includes individuals who were in the same county of those forced to relocate, but were not in the path of the dam construction. Robust standard errors in parentheses.

*** p<0.01, ** p<0.05, * p<0.1

Table A10. Effect of TVA relocation by home ownership, Georgia control group

VARIABLES	(1) Farm	(2) Employment	(3) Labor Force	(4) Unskilled Occupation	(5) Skilled Occupation
<u>A. Renter</u>					
Treat*Post	-0.098*** (0.019)	0.063** (0.024)	0.102*** (0.021)	0.089*** (0.025)	0.035 (0.023)
Observations	62,344	75,433	75,433	63,521	63,521
R-squared	0.063	0.013	0.037	0.033	0.089
Mean	0.544	0.627	0.610	0.354	0.280
<u>B. Owner</u>					
Treat*Post	0.006 (0.019)	0.032 (0.021)	0.032* (0.018)	0.016 (0.024)	-0.040** (0.016)
Observations	43,821	51,948	51,948	41,532	41,532
R-squared	0.031	0.012	0.035	0.042	0.040
Mean	0.624	0.568	0.537	0.352	0.259
Coefficients Different	No	Yes	Yes	Yes	Yes

Notes: Main effects for treatment and year fixed effects are included in each regression. Control group includes individuals living in Georgia who have-not-yet been treated (relocated) due to a later dam construction year. Robust standard errors in parentheses, *** p<0.01, ** p<0.05, * p<0.1

Table A11. Effect of TVA relocation by home ownership, matched county controls

VARIABLES	(1) Farm	(2) Employment	(3) Labor Force	(4) Unskilled Occupation	(5) Skilled Occupation
<u>A. Renter</u>					
Treat*Post	-0.241 (0.159)	0.072*** (0.023)	0.101*** (0.020)	0.088 (0.064)	0.014 (0.076)
Observations	3,776	3,103	3,103	2,633	2,633
R-squared	0.050	0.037	0.094	0.025	0.047
Mean	0.828	0.555	0.521	0.337	0.0831
<u>B. Owner</u>					
Treat*Post	-0.331 (0.204)	-0.034 (0.021)	0.019 (0.015)	0.054 (0.042)	-0.112*** (0.028)
Observations	2,677	1,986	1,986	1,634	1,634
R-squared	0.054	0.028	0.055	0.037	0.051
Mean	0.729	0.505	0.480	0.344	0.174
Coefficients Different	No	Yes	Yes	Yes	Yes

Notes: Main effects for treatment and year fixed effects are included in each regression. Control group includes individuals living in counties that were matched to treatment counties using a nearest neighbor matching algorithm based on 1930 county characteristics. Robust standard errors in parentheses,

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Table A12. Unskilled Occupations Breakdown

VARIABLES	(1) Unskilled (Overall)	(2) Unskilled without Farming	(3) Unskilled Farming
<u>A. Within-Counties Comparison</u>			
treat_post	0.045** (0.016)	0.060** (0.025)	-0.010 (0.010)
Observations	124,649	66,528	120,912
R-squared	0.003	0.013	0.013
Mean	0.323	0.144	0.584
<u>B. Georgia Have-Not-Yet Been Treated</u>			
treat_post	0.105*** (0.026)	0.124*** (0.036)	-0.083** (0.036)
Observations	4,267	1,803	4,375
R-squared	0.004	0.035	0.004
Mean	0.340	0.133	0.688
<u>C. Matched Counties Comparison</u>			
treat_post	0.066*** (0.021)	0.070** (0.028)	-0.011 (0.012)
Observations	105,053	57,049	101,271
R-squared	0.005	0.009	0.014
Mean	0.353	0.168	0.593

Notes: Main effects for treatment and year fixed effects are included in each regression. Column 1 consists of all unskilled jobs, which is the same variable used in our main analysis. Column 2 is similar to Column 1, but does not include any farm laborer jobs. Column 3 only consists of unskilled jobs related to farming such as farm laborers. Robust standard errors in parentheses, *** p<0.01, ** p<0.05, * p<0.1

VARIABLES	(1) Unskilled (Black)	(2) Unskilled Non-Farmer (Black)	(3) Unskilled Farmer (Black)	(4) Unskilled (White)	(5) Unskilled Non-Farmer (White)	(6) Unskilled Farmer (White)
<u>A. Within-Counties Comparison</u>						
Treat*Post	0.090 (0.061)	0.120* (0.064)	-0.014 (0.027)	0.038** (0.015)	0.053* (0.026)	-0.004 (0.007)
Observations	10,904	5,232	9,351	113,723	61,286	109,094
R-squared	0.022	0.070	0.005	0.003	0.010	0.017
Mean	0.481	0.361	0.674	0.308	0.125	0.576
<u>B. Georgia Have-Not-Yet Been Treated</u>						
Treat*Post	0.317*** (0.049)	0.235*** (0.048)	-0.062** (0.024)	0.083*** (0.025)	0.102** (0.046)	-0.028 (0.026)
Observations	438	155	413	3,822	1,643	3,837
R-squared	0.043	0.090	0.006	0.002	0.028	0.009
Mean	0.490	0.418	0.799	0.320	0.0994	0.674
<u>C. Matched Counties Comparison</u>						
Treat*Post	0.175*** (0.056)	0.189*** (0.058)	0.015 (0.027)	0.056** (0.021)	0.058** (0.028)	0.000 (0.008)
Observations	13,892	7,172	11,631	91,134	49,860	87,001
R-squared	0.010	0.031	0.024	0.008	0.010	0.017
Mean	0.514	0.403	0.668	0.328	0.132	0.583
A. Sig Difference by race?	Yes	Yes	No	Yes	Yes	No
B. Sig Difference by race?	Yes	Yes	Yes	Yes	Yes	Yes
C. Sig Difference by race?	Yes	Yes	No	Yes	Yes	No

Robust standard errors in parentheses

*** p<0.01, ** p<0.05, * p<0.1